

Certifying Through Repair

Decoder Selection and Finite-Horizon Representation Intervention

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September 2026

Integrated research draft with complete proofs and executable experiments

Abstract

Representation repair changes task loss and the evidence available for later decisions. We study two explicit contracts: fixed-confidence certification of the old representation, and finite-horizon Bayes control with a terminal declaration about the current representation. In an ordered single-channel experiment with observable environment-independent repair failures and free post-repair observations, we reduce old-label certification to choosing a decoder on post-repair observational classes. We prove its exact leading cost coefficient and a common attaining policy with fully charged deadline tails. Classwise factorization computes the coefficient and implements the likelihood-ratio race without exponential decoder enumeration. A finite-deadline information bound yields a matching shrinking-channel regime where distinct post laws fail to improve the leading coefficient. For current-label Bayes control, we characterize declaration-gate cost through Bellman advantages and bound repair-region components by the affine pieces of passive post-repair decision risk. A full-support TP2 example attains two disconnected repair regions and strict intervention below the declaration boundary. No-information and repeatable-information families give, respectively, an exact linear gate-cost example and a logarithmic upper bound. Rational Bayes computation, independent policy-tree enumeration, finite count-lattice evaluation, and complete source files accompany the results. The analysis uses established testing and POMDP tools; the stated contracts and structural assumptions delimit every claim.

Keywords: representation adequacy; repair; sequential testing; multiple correct answers; finite-horizon control; POMDP.

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1 Introduction

A representation changes which distinctions an agent can use. A repair can improve task performance, alter the label of the representation currently installed, and change the observations available for evaluating earlier decisions. These effects create two questions that should be studied together without identifying their objectives: how much evidence is needed to certify an old representation when repair is allowed, and when should an agent intervene under a finite task deadline?

For old-label certification, repairing first can change which pre-repair alternatives must be ruled out. Suppose three environments have old labels $(0, 0, 1)$, and a repair reveals whether the environment is in $\{1\}$ or $\{2, 3\}$. In environment 1, it may be efficient to rule out environment 2 and then use the decoder $(0, 1)$. The unresolved environment 3 has the opposite old label, but the post-repair observation can distinguish it from environment 1. Certifying the final old label before any repair would instead need to rule out environment 3 directly.

For current-label task control, the relevant decision can differ even without a high-confidence constraint. A Bayes declaration compares the two costs of misclassifying the installed representation. An intervention compares task losses, its own price, remaining time, and future information. These comparisons can favor different actions at the same belief. A noisy verification option can also split the optimal repair region into disconnected intervals.

We formalize both questions in finite known models with one joint observation–state convention and explicit loss accounting. All declared theorems include proofs. The principal results are:

1. A reduction from an ordered repair experiment to a finite multiple-correct-answer problem, with optimal old-label certification cost $g_i \log(1/\delta)/D_i$, a common attaining policy, and fully charged deadline tails (Theorem 4.2).
2. A factorization of D_i across post-repair classes and a polynomial eligibility test for the same likelihood-ratio policy (Theorems 5.1 and 5.2).
3. A finite-deadline information bound and a matching shrinking-channel regime in which the joint coefficient returns to the certification-before-repair coefficient, despite distinct post laws at every confidence level (Theorems 7.1 and 7.2).
4. For finite-horizon Bayes control, an exact declaration-gate cost identity, a bound on the number of repair-region components, and a full-support TP2 example attaining two components (Theorems 8.1, 9.1, and 9.3).
5. Exact no-information gate costs and a constructive logarithmic gate-cost upper bound with repeatable identifiable verification (Proposition 8.3 and Theorem 10.2), with reproducible finite computations under each stated contract.

The common message is precise: evidence required for a final assertion and evidence sufficient to select a useful intervention need not coincide. The two policy restrictions in this paper are different. Certification before repair commits to a fixed old-label answer under a uniform error constraint. A declaration gate restricts interventions according to a Bayes decision boundary about the current representation. We compare policies only within one objective and one set of physical kernels at a time.

Scope and contribution. The statistical max–min answer complexity, belief-state recursion, and information-comparison tools are established ideas. Our focus is their explicit repair interpretation, the classwise factorization, finite-deadline boundary, and repair-region structure in the stated models. No result assumes that calling a state a representation automatically makes a new control theory. Proposition 3.1 shows why arbitrary finite risk tables can fit the adequacy definition. Open-ended representation discovery, unknown model learning, and arbitrary repeated repair remain outside the proved scope.

2 Related work

The closest statistical comparison is pure exploration with multiple correct answers (Degenne and Koolen, 2019). Its answer-dependent alternatives become the invalidating environments of a post-repair decoder. With only one pre channel, there is no experiment-allocation optimization. The coefficient’s max–min form is therefore not claimed as a new principle. The ordered-process reduction and deadline accounting are established explicitly, while the disjoint class coordinates yield the polynomial implementation in Section 5.

Sequential change of measure connects sample counts and information (Kaufmann et al., 2016); non-uniform costs and controlled sensing have also been studied (Nitinawarat and Veeravalli, 2014). We use finite likelihood-ratio martingales and bounded-increment concentration (Hoeffding, 1963). The finite-deadline bound accounts for information from both phases and exposes a nonuniform limit of the free-post-information coefficient.

Finite-horizon POMDP belief states and piecewise-linear value functions are classical (Smallwood and Sondik, 1973). Their concavity supplies the repair-region component argument; the Bellman equation itself is not a contribution. Inspection and maintenance already combine observation and intervention (Morato et al., 2022; Zhang and Zhang, 2023). Structural POMDP results require conditions beyond an informative ordered channel (Krishnamurthy, 2018). Our TP2 example shows that this channel condition alone is insufficient for a connected repair region, and does not contradict theorems with additional cost and transition conditions. Pure-information monotonicity follows from Blackwell comparison (Blackwell, 1953).

Model editing provides a broader example of changing stored computational content (Meng et al., 2022), while learning to defer studies a predictor’s use of an expert (Mozannar and Sontag, 2020). These motivate repair and fallback, but we neither model a specific editing method nor claim empirical gains for a deployed language-model system. The fixed-representation M1 manuscript (Huang, 2026) motivates the adequacy question. The present proofs are self-contained; its fixed-confidence objective is not silently identified with a finite Bayes horizon.

Notation. All logarithms are natural. For finite laws, $\text{KL}(P\|Q) = \sum_y P(y) \log(P(y)/Q(y))$. We use $\min \emptyset = +\infty$ and $c/(+\infty) = 0$. A uniform error guarantee belongs to one implementable policy used in every environment. An environment-indexed infimum still ranges over this policy class; the controller is never given the true index.

3 Finite-horizon representation intervention

3.1 Environment, representation, and adequacy

Let $I \in \mathcal{E} = \{1, \dots, K\}$ be an unknown environment drawn once from a known prior μ . It remains fixed during deployment. Candidate representations form a finite set $\mathcal{F}$. In environment i , let H^{ref} be a finite reference task variable, let $\mathcal{U}$ be a finite task-action set, and let $\ell_i^{\text{ref}}(H^{\text{ref}}, u)$ be a nonnegative loss. Define

$$R_i^{*,\text{full}} = \min_d \mathbb{E}_i[\ell_i^{\text{ref}}(H^{\text{ref}}, d(H^{\text{ref}}))], \quad (1)$$

$$R_i^{*,f} = \min_{\tilde{d}} \mathbb{E}_i[\ell_i^{\text{ref}}(H^{\text{ref}}, \tilde{d}(f(H^{\text{ref}})))], \quad (2)$$

$$\Delta_i(f) = R_i^{*,f} - R_i^{*,\text{full}}, \quad \Theta_i(f) = \mathbf{1}\{\Delta_i(f) = 0\}. \quad (3)$$

Randomized rules can be included and do not improve these finite unconstrained minima. These are environment-wise reference optima, not a claim that the deployment controller knows i . When a stage task loss is identified with $\Delta_i(f)$, a realizable task/controller interface must justify that identification.

Proposition 3.1 (Finite risk-table realization). *For any nonnegative finite table $(d_{ij})_{i \leq K, j \leq M}$, there is a finite reference problem and M representations satisfying $R_i^{*,\text{full}} = 0$ and $\Delta_i(f_j) = d_{ij}$. Their visible representation outputs can have the same distribution in every environment.*

Proof. Take independent fair bits $Z_1, \dots, Z_M$, task action $u \in \{0, 1\}^M$, and loss $\ell_i^{\text{ref}}(Z, u) = 2 \sum_j d_{ij} \mathbf{1}\{u_j \neq Z_j\}$. Let f_j expose every bit except Z_j . Full information gives zero loss. Under f_j , copying observed bits and guessing the missing bit incurs d_{ij} ; independence makes a lower loss impossible. The joint bit distribution does not depend on i . In deployment, use fresh bits each round and do not reveal task targets or actual task losses beyond this interface. $\square$

This proposition also limits the semantics: arbitrary nonnegative loss tables can be represented in this way. Merely assigning adequacy labels to a finite control problem therefore adds no automatic structural restriction.

3.2 Visible state, actions, and event order

The visible state is $X_t = (F_t, M_t, B_t)$ in a finite set $\mathcal{X}$, comprising a representation, policy mode, and intervention budget. At time $t = 0, \dots, T - 1$, the agent chooses $A_t \in \mathcal{A}(X_t)$, a nonempty finite set. The available actions may include KEEP, VERIFY, REPAIR, FALLBACK, DEFER, and additional known information actions. A fallback changes the policy mode without changing the representation; repair changes the representation and can consume a budget even on failure.

The observable event is the pair (Y_{t+1}, X_{t+1}) with joint kernel

$$\mathbf{K}_i(y, x' \mid x, a), \quad \sum_{y, x'} \mathbf{K}_i(y, x' \mid x, a) = 1. \quad (4)$$

The loss is $\ell_i(x, a, y, x') \in [0, L_{\max}]$, including task and direct action costs exactly once. Loss is not implicitly observed. If reward, task success, or actual loss is visible, that feedback must be included in the kernel. In particular, a visible successful repair may itself reveal information when its probability depends on the environment.

The general kernel permits either intervention timing convention if the loss is specified consistently. All analytic binary instances and all supplied family configurations use *intervention before the current task*: a successful repair affects the current round as well as subsequent rounds. A failed attempt uses the old representation in its current round. Verification costs and any task loss during verification are explicit configuration parameters. The separate three-environment fixture in Section 11 instead uses next-task repair; its costs are explicitly different.

3.3 Terminal decisions and a common objective

At time T , the agent selects $D_T \in \mathcal{D}(X_T)$, where binary decisions 0, 1 are always available and abstention may additionally be available. The terminal loss is

$$\begin{aligned} L_i(x, d) = & L_i^{\text{deploy}}(x) + \lambda_{\text{miss}} \mathbf{1}\{\Theta_i(f_x) = 0, d = 1\} \\ & + \lambda_{\text{FA}} \mathbf{1}\{\Theta_i(f_x) = 1, d = 0\} + \lambda_{\perp} \mathbf{1}\{d = \perp\}. \end{aligned} \quad (5)$$

All terminal losses are finite and nonnegative. The objective is

$$J_T(\pi; \mu, x_0) = \mathbb{E}_{\mu}^{\pi} \left[\sum_{t=0}^{T-1} \ell_I(X_t, A_t, Y_{t+1}, X_{t+1}) + L_I(X_T, D_T) \right]. \quad (6)$$

There is no intermediate commitment action and no cancellation of task losses after a declaration. The terminal target is $\Theta_I(F_T)$, not $\Theta_I(F_0)$.

3.4 Belief recursion and the declaration benchmark

For the observable history $\mathcal{H}_t$, let $b_t(i) = \mathbb{P}(I = i \mid \mathcal{H}_t)$. The predictive probability and posterior are

$$q_b(y, x' \mid x, a) = \sum_i b(i) \mathsf{K}_i(y, x' \mid x, a), \quad (7)$$

$$\Phi_i(b, x, a, y, x') = \frac{b(i) \mathsf{K}_i(y, x' \mid x, a)}{q_b(y, x' \mid x, a)}. \quad (8)$$

Only events with positive predictive probability are conditioned upon. Write $p(b, x) = \sum_i b(i) \mathbf{1}\{\Theta_i(f_x) = 0\}$. With binary declarations, positive error costs, and no abstention, an immediate declaration would prefer *inadequate* exactly when

$$p(b, x) \geq p_{\text{dec}} = \frac{\lambda_{\text{FA}}}{\lambda_{\text{FA}} + \lambda_{\text{miss}}}. \quad (9)$$

Equality permits declaring inadequate. This is a Bayes classification boundary. For symmetric costs it is $1/2$; it is not $1 - \delta$ and is not a high-confidence certification requirement.

Define $G_h(b, x) \subseteq \mathcal{A}(x)$ by removing REPAIR and FALLBACK whenever $p(b, x) < p_{\text{dec}}$, with other actions retained. We assume $G_h(b, x)$ remains nonempty. The joint and gated policy classes use the same model and objective; only this restriction differs. An abstention extension requires a separately specified gate based on its full terminal decision regions.

3.5 Standard optimal control

With h rounds left, write $V_h(b, x)$ for the unrestricted minimum loss. Then

$$V_0(b, x) = \min_d \sum_i b(i) L_i(x, d), \quad (10)$$

$$\mathcal{Q}_h(b, x, a) = \sum_{y, x'} \sum_i b(i) \mathsf{K}_i(y, x' \mid x, a) [\ell_i(x, a, y, x') + V_{h-1}(\Phi(b, x, a, y, x'), x')], \quad (11)$$

$$V_h(b, x) = \min_{a \in \mathcal{A}(x)} \mathcal{Q}_h(b, x, a). \quad (12)$$

Zero-probability terms are omitted. The continuation value inside the sum is independent of the summation index i .

Proposition 3.2 (Standard sufficiency and concavity). *The state (h, b, x) is sufficient. A deterministic belief-Markov optimum exists. For fixed h, x , $V_h(b, x)$ is the minimum of finitely many linear functions of b , hence is continuous, piecewise linear, and concave.*

Proof. Conditioning on (b, x) determines the joint law of each next event and its expected cost. Backward induction proves sufficiency and attainment over finite action sets. Alternatively, fix any deterministic observation-contingent policy tree, including its terminal declarations. Its environment-wise losses form a vector α , and its Bayes loss is $b^\top \alpha$. There are finitely many such trees for finite horizon and alphabets. Minimizing over these vectors proves the claimed geometry. $\square$

These properties hold for the unrestricted model. A gate whose admissible actions depend on belief need not preserve concavity. Also, for $K > 2$, the scalar $p(b, x)$ generally does not determine kernels or future losses; the entire belief is needed.

3.6 The two contracts used in this paper

The joint-event convention applies to both parts, but their objectives are specified separately. The certification part fixes the target $\theta_i = \Theta_i(F_0)$ and imposes a uniform error constraint at a confidence-dependent deadline. The Bayes part uses (6) and the actual terminal representation F_T . Certification pre-slot costs g_i include observation and task costs; they are not stipulated to equal the excess risk $\Delta_i(F_0)$. The zero post-cost condition is a substantive additional assumption.

	Old-label certification	Current-label control
Target	$\Theta_i(F_0)$	$\Theta_i(F_T)$
Error treatment	Uniform probability $\leq \delta$	Terminal Bayes penalty
Time	$H_\delta = \lceil \log^2(1/\delta) \rceil$	Fixed finite T
Restriction	Commit to old answer before re-pair	Intervention gate at stated posterior
Post information	Free in cost, consumes slots	Specified joint kernel and costs

Every policy comparison preserves its row’s target, costs, and kernels. Positive integer macro durations with delayed completion feedback can be expanded into unit slots as proved in Appendix I.

4 Old-label certification through one repair

4.1 The ordered experiment

We now impose a narrower experiment than the control model above. Let $\theta_i \in \{0, 1\}$ denote adequacy of the *old* representation. It remains the terminal target after repair.

Assumption 4.1 (Finite ordered repair experiment). The following parameters are fixed as $\delta \downarrow 0$.

1. There are finitely many environments. Pre-repair observations are iid with pairwise distinct laws P_i on a common finite alphabet, all with full support. The binary labels include both values.
2. A pre-repair task slot costs $g_i > 0$ and can supply one observation from P_i . There is no free informative pre-repair action. Pausing or completing old-state tasks does not remove their cost.
3. After entering the repair phase, no more pre-repair observations are available. Each attempt takes one slot and costs $\kappa_i \geq 0$, including its opportunity cost. It succeeds with known probability $q > 0$, independent of the environment, previous observations, and other attempts. Success is observable and irreversible. A failed attempt can be retried.
4. After success, each slot supplies an iid observation with law Q_i on a finite alphabet at zero cost. These observations are independent of the pre-repair stream and repair trials conditional on i . Distinct Q_i need not exist for every pair. For simplicity all Q_i have full support.
5. All action choices use observed data and independent policy randomization. Accounting costs are not additional signals. All slots up to the prescribed deadline are accounted for, and a declaration is produced by that deadline.

The cost κ_i is the total cost of a repair slot, not an extra fee to which an omitted old-state loss must later be added. Post-repair observations are free in cost, not in time. Policies may stop trying repair, in which case remaining old-state task slots incur g_i . A policy that never

repairs has only a direct pre-repair declaration. These conventions exclude uncharged tails while retaining the natural direct-certification alternative.

Let $L_\delta = \log(1/\delta)$ and

$$H_\delta = \lceil L_\delta^2 \rceil. \quad (13)$$

For sufficiently small δ , let Π_δ be the ordered policies with $\mathbb{P}_i^\pi(\hat{\theta} \neq \theta_i) \leq \delta$ for every i , and define

$$J_i^*(\delta) = \inf_{\pi \in \Pi_\delta} \mathbb{E}_i^\pi C.$$

The separated class Π_δ^{sep} additionally commits to its final old-label answer *before its first repair attempt*, using only pre-repair observations and policy coins; it must meet the same uniform error guarantee. A no-repair policy commits by the deadline. Define $J_i^{\text{sep}}(\delta)$ by the corresponding infimum. This is an operational certification-before-repair restriction, not a Bayes posterior threshold.

4.2 Post-repair classes and admissible answers

Let $\mathcal{C}$ be the partition induced by exact equality of the post-repair laws, and write C_i for the class containing i . A decoder is a map $a : \mathcal{C} \rightarrow \{0, 1\}$. Define

$$\mathcal{D}_i = \{a : a(C_i) = \theta_i\}, \quad B(a) = \{j : a(C_j) \neq \theta_j\}. \quad (14)$$

The decoder is valid in environment i exactly when $a \in \mathcal{D}_i$. Its invalidating environments may have either old label. For a valid decoder, set

$$d_i(a) = \min_{j \in B(a)} \text{KL}(P_i \| P_j), \quad (15)$$

$$D_i = \max_{a \in \mathcal{D}_i} d_i(a), \quad D_i^{\text{sep}} = \min_{j: \theta_j \neq \theta_i} \text{KL}(P_i \| P_j). \quad (16)$$

There are only $2^{|\mathcal{C}|}$ decoders. No optimization over observation allocation appears because there is exactly one pre-repair channel.

Theorem 4.2 (Decoder cost coefficient). *Under Assumption 4.1, as $\delta \downarrow 0$,*

$$\frac{J_i^*(\delta)}{L_\delta} \longrightarrow \frac{g_i}{D_i}, \quad \frac{J_i^{\text{sep}}(\delta)}{L_\delta} \longrightarrow \frac{g_i}{D_i^{\text{sep}}} \quad \text{for every } i. \quad (17)$$

A single sequence of implementable joint policies attains the joint coefficients simultaneously in all environments. If a decoder is valid in every environment, the joint coefficient is zero and a policy of uniformly bounded expected cost exists for all sufficiently small δ .

Because the constant decoder $a(C) = \theta_i$ is valid in i , $D_i \geq D_i^{\text{sep}}$ and joint repair cannot have a larger leading coefficient. This comparison does not say that a finite- δ cost ratio equals the asymptotic coefficient ratio.

4.3 Three environments and same-label exclusion

Take Bernoulli laws with

$$(p_1, p_2, p_3) = (0.5, 0.9, 0.6), \quad (q_1, q_2, q_3) = (0.8, 0.2, 0.2), \quad (\theta_1, \theta_2, \theta_3) = (0, 0, 1), \quad g_i = 1. \quad (18)$$

Here q_i denotes a post-repair mean, whereas q without a subscript denotes the independent repair success probability. The classes are $C_A = \{1\}$ and $C_B = \{2, 3\}$. Listing decoders by $(a(C_A), a(C_B))$ gives

Decoder	$B(a)$	Valid environments
$(0, 0)$	$\{3\}$	1, 2
$(0, 1)$	$\{2\}$	1, 3
$(1, 0)$	$\{1, 3\}$	2
$(1, 1)$	$\{1, 2\}$	3

Under E_1 , the best decoder is $(0, 1)$. It excludes E_2 , which shares E_1 's old label, while retaining the opposite-label E_3 for post-repair discrimination. Direct old-label certification instead must exclude E_3 . For Bernoulli means,

$$\text{KL}(\text{Bern}(p) \parallel \text{Bern}(r)) = p \log(p/r) + (1 - p) \log((1 - p)/(1 - r)).$$

Thus

$$\begin{aligned} D_1 &= \text{KL}(\text{Bern}(0.5) \parallel \text{Bern}(0.9)) = 0.510825623766 \dots, \\ D_1^{\text{sep}} &= \text{KL}(\text{Bern}(0.5) \parallel \text{Bern}(0.6)) = 0.020410997260 \dots, \\ \frac{g_1}{D_1} &= 1.957615188971 \dots, \quad \frac{g_1}{D_1^{\text{sep}}} = 48.993196523204 \dots, \\ \frac{J_1^{\text{sep}}(\delta)}{J_1^*(\delta)} &\longrightarrow 25.026980174255 \dots \end{aligned}$$

The other environments and their best decoders are calculated in Table 1. These values are generated from the model file.

Environment	D_i	g_i/D_i	g_i/D_i^{sep}	Decoder
E_1	0.510826	1.957615	48.993197	$(0, 1)$
E_2	0.226289	4.419125	4.419125	$(0, 0)$
E_3	0.311239	3.212968	49.663496	$(0, 1)$

Table 1: Analytic decoder coefficients for (18). Costs have $g_i = 1$. Decoder coordinates follow (C_A, C_B) .

4.4 Partition boundaries

Corollary 4.3 (Preservation, erasure, and refinement). *Under the theorem's assumptions:*

1. *If every post-repair class has a constant old label, a universally valid decoder exists, $D_i = +\infty$, and bounded expected joint cost is achievable.*
2. *If all post-repair laws are equal, $D_i = D_i^{\text{sep}}$ for every i .*
3. *Refining the post-repair partition while keeping the pre-repair experiment and labels fixed cannot decrease D_i .*

Proof. For the first statement, assign each class its common label. No environment invalidates that decoder. For the second, a decoder on one class is a constant label, so its invalidating set is exactly the opposite-label set. For the third, lift any coarse decoder to a finer partition by assigning the parent class's value to each child. Its invalidating set stays unchanged, while the finer partition may allow additional decoders. Maximizing over that larger set cannot reduce the rate. $\square$

Partition equality here is exact. Two nearly equal but distinct post laws remain separate classes in the asymptotic theorem, although distinguishing them at practical deadlines can be expensive in time.

5 Decoder structure and polynomial implementation

The answer space contains 2^M maps, where $M = |\mathcal{C}|$. Its coordinates are indexed by disjoint observational classes, which gives a factorization that is not available for an arbitrary multiple-answer identification problem.

5.1 Computing the coefficient without enumerating answers

For a class C and a candidate output $\ell \in \{0, 1\}$, put

$$r_i(C, \ell) = \min_{j \in \mathcal{C}: \theta_j \neq \ell} \text{KL}(P_i \| P_j), \quad \mathcal{L}_i(C) = \begin{cases} \{\theta_i\}, & C = C_i, \\ \{0, 1\}, & C \neq C_i. \end{cases}$$

The empty minimum is $+\infty$.

Theorem 5.1 (Factorized decoder coefficient). *Under Assumption 4.1,*

$$D_i = \min_{C \in \mathcal{C}} \max_{\ell \in \mathcal{L}_i(C)} r_i(C, \ell). \quad (19)$$

An optimal decoder can be obtained by selecting, independently in each class, any allowed ℓ with $r_i(C, \ell) \geq D_i$. Choosing the smallest such ℓ in every coordinate gives the lexicographically first global optimum. Given the pairwise KL matrix, all D_i and a corresponding decoder for each environment require $O(K^2 + KM)$ arithmetic operations.

Proof. For every decoder valid in i ,

$$d_i(a) = \min_{C \in \mathcal{C}} r_i(C, a(C)).$$

Its coordinate in class C belongs to $\mathcal{L}_i(C)$. Thus $d_i(a) \leq \min_C \max_{\ell \in \mathcal{L}_i(C)} r_i(C, \ell)$. Conversely choose a maximizing label in each class. All choices satisfy validity in i , and their minimum equals the displayed upper bound. This proves equality. Once D_i is known, a valid decoder is globally optimal exactly when every coordinate meets this threshold. Coordinatewise smallest feasible choices therefore yield the first lexicographic optimum; requiring every coordinate to be a local maximizer would unnecessarily discard some global optima.

For one row of the KL matrix, scan the K environments and keep, in each class, the smallest divergence for each of the two labels. These give every $r_i(C, \ell)$ in $O(K + M)$ operations. Scan the classes again to take the bottleneck and construct a decoder. Repeat for all i . Computing the KL matrix itself uses $O(K^2 |\mathcal{X}_{\text{pre}}|)$ operations for a finite pre-repair alphabet, before accounting for numerical precision. $\square$

For a class containing only one old label, choosing that label incurs no invalidating alternative and contributes $+\infty$ to the bottleneck. Mixed-label classes are therefore the only effective constraints. In class C_i the output is fixed to θ_i ; other classes may be assigned whichever label leaves the easier set of alternatives to exclude. This is the mechanism behind same-label exclusion in Section 4.3.

5.2 A surviving-set test for the likelihood-ratio race

Let $s_j(n) = \sum_{t=1}^n \log P_j(X_t)$ and define, for a candidate witness k ,

$$S_k(n) = \{j : s_k(n) - s_j(n) < \beta_\delta\}.$$

It includes k , since $\beta_\delta > 0$.

Theorem 5.2 (Polynomial eligibility test). *A decoder eligible with witness k in (22) exists if and only if every nonempty intersection $S_k(n) \cap C$ has a single old label. For such a witness, each nonempty intersection forces that label as the coordinate $a(C)$; every empty intersection leaves the coordinate free. Consequently, the lexicographically first eligible decoder can be found in $O(K^2 + KM)$ operations per observation, without enumerating 2^M maps. For the separated race, eligibility is equivalent to all members of $S_k(n)$ having one common old label.*

Proof. The likelihood condition requires every invalidating environment $j \in B(a)$ to lie outside $S_k(n)$. Equivalently, $a(C_j) = \theta_j$ for every surviving j . One coordinate can satisfy these requirements precisely when the labels within each surviving class intersection agree. Since k survives, the resulting decoder is automatically valid at its witness.

For each witness, assign the forced labels and zero to unconstrained coordinates. This is the smallest eligible decoder for that witness. Taking the smallest of these candidates, with the witness index as a secondary tie-break, returns exactly the first decoder of an exhaustive lexicographic race. A scan over the K environments followed by the M coordinates suffices for each witness. Constant decoders satisfy all surviving constraints exactly when their labels agree globally. $\square$

This implementation realizes the same statistical rule analyzed below; it does not change its threshold or error guarantee. An online controller stores K log-likelihood scores, the class and label maps, and a candidate decoder. Its working storage can be $O(K + M)$ when witnesses are checked successively. Floating-point equality at a likelihood threshold is still a numerical issue: the exact rule uses the displayed strict surviving-set inequality, and the implementation uses the complementary comparisons consistently.

The finite Bernoulli evaluator batches many possible count histories, so its time and storage costs differ from those of a single online trajectory. Its bit-coded batch implementation supports at most 60 classes; the scalar polynomial implementation has no such encoding restriction.

6 Proof of the decoder coefficient

6.1 Reducing a repair policy to a random decoder

Lemma 6.1 (Decoder reduction). *For any ordered policy in Π_δ , one can construct, on an enlarged probability space, its number N of pre-repair observations and a random decoder A such that*

$$\mathbb{P}_j(A \notin \mathcal{D}_j) \leq \delta \quad \text{for every } j. \quad (20)$$

There is auxiliary randomness U , whose distribution is independent of the environment and of the entire pre-repair stream, for which N is a stopping time in the filtration $\sigma(U, X_1, \dots, X_n)$ and A is measurable at N . The expected total cost is at least $g_i \mathbb{E}_i N$. For a separated policy, A can be taken to be a constant decoder.

Proof. Pre-generate an infinite iid pre-repair stream, including observations unused by the physical policy. Put policy coins, repair-trial coins, and independent uniform variables for each possible post-repair class into U . Their joint distribution does not depend on the true environment. Uniform variables can generate a sample stream from the known law Q_C for each class C .

Follow the policy until it leaves the pre-repair phase, or makes a no-repair terminal decision. Given the pre-repair history and U , its last pre-repair sample count is a stopping time: no post-repair observation can affect a later decision to take another pre-repair sample. In the no-repair case, set every coordinate of A to the final declaration. Otherwise, for each class C , simulate the policy's remaining repair and post-repair continuation using the same pre-repair internal state, the repair coins, and the synthetic Q_C stream; let $A(C)$ be its final label. A

continuation that never succeeds in repair still produces a deadline declaration under the policy's rule. The simulations need not be available to the physical policy. They define an auxiliary statistical answer for the lower bound.

Under environment j , the marginal simulated continuation for C_j has exactly the original policy's continuation law. In particular, its failed repair trials, any abort decision, and its deadline rule have the correct distribution. Thus the event $A(C_j) \neq \theta_j$ has the original error probability, proving (20). The other coordinates only supply a coupling. Both N and A are functions of the pre-repair history and U , and A is measurable at N in the enlarged filtration. A separated policy's already committed label defines a constant decoder. Every pre-repair observation costs at least g_i , and all other costs are nonnegative, yielding the cost bound. $\square$

The environment-independent repair mechanism is essential to this reduction. If success probabilities depend on i , their likelihood ratios are extra information, and the same coefficient need not apply. Likewise, using unobserved accounting costs as signals would invalidate the experiment.

6.2 A lower bound with multiple correct answers

Lemma 6.2 (Pre-repair sample lower bound). *If $D_i < \infty$, uniformly over uniformly δ -correct ordered policies,*

$$\liminf_{\delta \downarrow 0} \frac{\mathbb{E}_i N}{L_\delta} \geq \frac{1}{D_i}.$$

For separated policies the same bound holds with D_i^{sep} .

Proof. Fix i and $0 < \varepsilon < 1$. Every valid decoder has a nonempty invalidating set when $D_i < \infty$. For each $a \in \mathcal{D}_i$, choose $j_a \in B(a)$ attaining $d_i(a)$. Put

$$n = \left\lfloor \frac{(1 - \varepsilon)L_\delta}{D_i} \right\rfloor, \quad u = (1 - \varepsilon/2)L_\delta, \quad L_{ij}(n) = \sum_{t=1}^n \log \frac{P_i(X_t)}{P_j(X_t)}.$$

The event $F_a = \{N \leq n, A = a\}$ is measurable in $\sigma(U, X_1, \dots, X_n)$. On this fixed-time sigma-field the density ratio between environments i and j_a is $\exp L_{ij_a}(n)$, since U has the same law and is independent of the pre-repair stream. Therefore

$$\begin{aligned} \mathbb{P}_i(F_a) &\leq e^u \mathbb{P}_{j_a}(F_a) + \mathbb{P}_i\{L_{ij_a}(n) > u\} \\ &\leq \delta^{\varepsilon/2} + \mathbb{P}_i\{L_{ij_a}(n) > u\}. \end{aligned} \tag{21}$$

The second inequality uses $j_a \in B(a)$ and (20). The increments are bounded and iid under i , and

$$\mathbb{E}_i L_{ij_a}(n) = n d_i(a) \leq n D_i \leq (1 - \varepsilon)L_\delta.$$

The gap to u is at least $\varepsilon L_\delta/2$. A bounded-increment concentration bound makes the remaining probability $o(1)$, independently of the policy. There are finitely many decoders. Adding (21) over valid a and using $\mathbb{P}_i(A \notin \mathcal{D}_i) \leq \delta$ gives $\mathbb{P}_i(N \leq n) = o(1)$. Consequently $\mathbb{E}_i N \geq n \mathbb{P}_i(N > n) = n(1 - o(1))$. First let $\delta \downarrow 0$, then $\varepsilon \downarrow 0$.

A separated policy permits only constant decoders. The unique constant decoder valid in i has invalidating set $\{j : \theta_j \neq \theta_i\}$. Repeating the argument gives the separated rate. $\square$

The proof deliberately uses a fixed-time likelihood ratio on the extended iid stream. It does not substitute an unstopped likelihood for a stopped one, and it does not lower-bound each possible answer as if that answer had to be selected with probability approaching one.

6.3 A common likelihood-ratio decoder procedure

At pre-repair sample count n , define the stopping rule

$$T = \inf \{n \geq 0 : \exists a, \exists k \notin B(a), L_{kj}(n) \geq \beta_\delta \text{ for all } j \in B(a)\}, \quad \beta_\delta = \log \frac{4K(K-1)}{\delta}. \quad (22)$$

Empty invalidating sets satisfy the condition at $n = 0$. A deterministic ordering resolves simultaneous eligible decoders and witnesses. The witness k is searched over all environments; it is not the unknown true index supplied to the algorithm.

Lemma 6.3 (Error and stopping time of the race). *The probability that the race ever selects an invalid decoder in environment i is at most $\delta/4$. If $D_i < \infty$, then*

$$\limsup_{\delta \downarrow 0} \frac{\mathbb{E}_i T}{L_\delta} \leq \frac{1}{D_i}.$$

Moreover, for $m = \lfloor H_\delta/3 \rfloor$ and sufficiently small δ , $\mathbb{P}_i(T > m) \leq A_i e^{-a_i m}$ with fixed positive constants A_i, a_i .

Proof. Under i , an invalid selected decoder must have a witness $k \neq i$ with $L_{ki}(n) \geq \beta_\delta$. The process $\exp L_{ki}(n)$ is a nonnegative unit-mean likelihood-ratio martingale under i . Ville's maximal inequality and a union bound give

$$\mathbb{P}_i(\text{invalid decoder ever selected}) \leq (K-1)e^{-\beta_\delta} \leq \delta/4.$$

No union over all decoders is needed.

Choose an optimal $a_i \in \mathcal{D}_i$ and set $T_i = \inf\{n : \min_{j \in B(a_i)} L_{ij}(n) \geq \beta_\delta\}$. The race satisfies $T \leq T_i$. For a fixed $\eta > 0$, let $n_0 = \lceil (1+\eta)\beta_\delta/D_i \rceil$. For $n \geq n_0$, every $j \in B(a_i)$ has drift $d_{ij} \geq D_i$, and

$$nd_{ij} - \beta_\delta \geq nD_i \frac{\eta}{1+\eta}.$$

If $T_i > n$, at least one likelihood ratio is below threshold at time n . Bounded-increment concentration therefore gives, for some $c_{i,\eta} > 0$,

$$\mathbb{P}_i(T > n) \leq \mathbb{P}_i(T_i > n) \leq |B(a_i)|e^{-c_{i,\eta}n} \quad (n \geq n_0).$$

Summing tail probabilities bounds $\mathbb{E}_i T$ by $n_0 + o(1)$ as $\delta \downarrow 0$. Divide by L_δ , then let $\eta \downarrow 0$. For the deadline tail use a fixed η and $m/\beta_\delta \rightarrow \infty$ to obtain the stated exponential bound. If a universal decoder exists, $T = 0$ and both claims are immediate. $\square$

6.4 A deadline policy with fully charged tails

Set $m = \lfloor H_\delta/3 \rfloor$. Run the race until it selects a decoder or until m pre-repair observations have been taken. If no decoder is selected, finish all remaining slots in the old state and declare any fixed pre-history rule, such as its maximum-likelihood label. This event is conservatively counted as error. Otherwise attempt repair for at most m slots. If all attempts fail, finish the remaining old-state task slots, charge their costs, and declare a pre-history rule. If repair succeeds, use the remaining post-repair observations to choose a maximum-likelihood post class $\hat{C}$, and output the selected decoder's label $a(\hat{C})$. There are at least m post observations after any such success. For a separated policy, restrict the race to constant decoders and retain its committed answer even on the all-failed branch.

For distinct post classes C, D , let $\rho_{CD} = \sum_y \sqrt{Q_C(y)Q_D(y)} < 1$ and $\rho = \max_{C \neq D} \rho_{CD}$. Under Q_C , for any r observations, Markov's inequality applied to the square root of the likelihood ratio yields

$$\mathbb{P}_C \left\{ \prod_{t=1}^r Q_D(Y_t) \geq \prod_{t=1}^r Q_C(Y_t) \right\} \leq \rho_{CD}^r.$$

Thus maximum-likelihood classification among $M = |\mathcal{C}|$ classes has error at most $(M - 1)\rho^m$ when $r \geq m$. Take this bound as zero for $M = 1$.

The total terminal error is bounded by

$$\frac{\delta}{4} + \mathbb{P}_i(T > m) + (1 - q)^m + (M - 1)\rho^m. \quad (23)$$

The last three terms are $o(\delta)$, because all instance parameters are fixed and m grows as $L_\delta^2/3$. Hence the policy belongs to Π_δ for all sufficiently small δ . For separated policies the post-class term is unnecessary; the pre-declaration error is already bounded by $\delta/4 + \mathbb{P}_i(T > m)$.

The expected number of repair attempts on any eligible history is at most $1/q$. Charging timeout and all-failed tails by the entire possible remaining old-state cost gives the upper bound

$$\mathbb{E}_i C \leq g_i \mathbb{E}_i(T \wedge m) + \frac{\kappa_i}{q} + g_i H_\delta [\mathbb{P}_i(T > m) + (1 - q)^m]. \quad (24)$$

This bound includes both failed attempts and unrepaired task slots. Its last term is $o(1)$. Lemma 6.3 gives the desired upper coefficient. Lemmas 6.1 and 6.2 give the lower coefficient. Restricting to constant decoders proves the separated equality. If a universal decoder exists, $T = 0$ and (24) is bounded by $\kappa_i/q + o(1)$. This proves Theorem 4.2.

Remark 6.4 (The role of the horizon). The square-logarithmic deadline is a sufficient explicit schedule. It makes all exponentially small tail terms negligible relative to δ and their task costs negligible relative to L_δ . It is not claimed to be the smallest feasible deadline. A fixed horizon with bounded likelihood ratios cannot support arbitrarily small uniform error for all these instances.

7 Finite deadlines and post-repair information

7.1 An explicit finite protocol

Fix integers N, R, H with $N + R < H$. The asymptotic policy uses $N = R = \lfloor H_\delta/3 \rfloor$, but a finite experiment must specify its budgets explicitly. Run the polynomial race of Theorem 5.2 as follows.

```

For n = 0, ..., N:
    Check decoder eligibility using the first n samples.
    If eligible, select the first decoder and leave this loop.
    If n = N, charge the old-state tail and declare a fallback label.
    Otherwise collect one more pre-repair observation.
After selecting a decoder:
    Attempt repair at most R times, stopping on success.
    On success, classify the post class and apply the decoder.
    On all failures, charge the old-state tail and declare fallback.
    
```

The boundary sample $n = N$ is checked before timeout. In the separated version the selected decoder is constant and its label is retained on every subsequent branch, including failed repair.

Let T be the untruncated race time and $f = (1 - q)^R$. The same argument as in Section 6 gives the finite error bound

$$\mathbb{P}_i(\hat{\theta} \neq \theta_i) \leq \frac{\delta}{4} + \mathbb{P}_i(T > N) + f + (M - 1)\rho^{H-N-R}, \quad (25)$$

when maximum-likelihood post classification is used. The last term is omitted for one class. It also gives

$$\mathbb{E}_i C \leq g_i \mathbb{E}_i(T \wedge N) + \frac{\kappa_i(1 - f)}{q} + g_i H \{\mathbb{P}_i(T > N) + f\}. \quad (26)$$

These are unconditional bounds. In particular, a path-specific sample count must not be mixed with an unconditional timeout probability and called a conditional expectation. A positive threshold parameter alone does not imply that arbitrary finite budgets achieve error δ .

For Bernoulli post means, the implementation may instead use nearest-mean classification. With minimum distinct-mean gap Δ_Q , its last error term can be replaced by $2 \exp(-(H - N - R)\Delta_Q^2/2)$, by the bounded-sum inequality (Hoeffding, 1963). Both classifiers coincide for the symmetric two-class means 0.8 and 0.2 used in the main finite experiment.

7.2 A quantitative information bound before taking a limit

The exact equality partition describes what can eventually be distinguished. At a finite deadline, the divergences between post laws also matter. Let N_P, N_Q be the numbers of acquired pre- and post-repair observations, respectively, and let

$$h_\delta = \text{kl}(1 - \delta, \delta) = (1 - 2\delta) \log \frac{1 - \delta}{\delta}, \quad 0 < \delta < 1/2.$$

Theorem 7.1 (Finite-deadline information lower bound). *For every uniformly δ -correct policy in the ordered experiment with deadline H , and every opposite-old-label pair i, j ,*

$$\mathbb{E}_i N_P \text{KL}(P_i \| P_j) + \mathbb{E}_i N_Q \text{KL}(Q_i \| Q_j) \geq h_\delta. \quad (27)$$

In particular,

$$J_i^*(\delta; H) \geq g_i \max_{j: \theta_j \neq \theta_i} \frac{[h_\delta - H \text{KL}(Q_i \| Q_j)]_+}{\text{KL}(P_i \| P_j)}. \quad (28)$$

The latter is a lower bound, not a general matching coefficient.

Proof. Include the policy randomization in the complete transcript. It has the same distribution under both environments. Conditional on an observed history, the policy uses the same action kernel; each acquired pre observation contributes $\text{KL}(P_i \| P_j)$ to the expected log-likelihood ratio, and each post observation contributes $\text{KL}(Q_i \| Q_j)$. Repair-success trials, deterministic time advancement, and environment-independent policy coins contribute zero. The finite-horizon chain rule therefore gives the left side of (27) as the transcript divergence. Ignoring an available observation can only reduce what the terminal rule uses, and does not invalidate the bound.

Map the transcript to the event that the terminal answer equals θ_i . Under i this event has probability at least $1 - \delta$; under the opposite-label j it has probability at most δ . Data processing for relative entropy and monotonicity of binary divergence in these two ranges give the lower bound h_δ . This is the standard sequential change-of-measure principle (Kaufmann et al., 2016), applied to the two phases. Finally $N_Q \leq H$, $C \geq g_i N_P$, and nonnegativity give (28). Take the infimum over uniformly correct policies. $\square$

This bound deliberately uses opposite-label alternatives because it concerns the *final old answer*. It need not recover the sharper decoder bound at a fixed post experiment. Their purposes differ: (28) quantifies a finite information budget, while Theorem 4.2 exploits which answer can be selected before a fixed post experiment.

7.3 A shrinking-channel limit that removes the coefficient advantage

The fixed-instance qualification in Theorem 4.2 cannot be dropped. Allow only the post laws to vary with δ , writing Q_i^δ , while the pre laws, labels, costs, and $q > 0$ remain fixed.

Theorem 7.2 (Collapse to the separated coefficient). *Let $H_\delta = \lceil L_\delta^2 \rceil$. If, for every opposite-label pair,*

$$H_\delta \text{KL}(Q_i^\delta \| Q_j^\delta) = o(L_\delta), \quad (29)$$

then the optimal joint cost for this sequence of experiments satisfies

$$\lim_{\delta \downarrow 0} \frac{J_i^*(\delta; Q^\delta)}{L_\delta} = \frac{g_i}{D_i^{\text{sep}}} \quad \text{for every } i. \quad (30)$$

Thus post laws can be pairwise distinct at every confidence level, yet give no improvement in the leading coefficient along the sequence.

Proof. Since $h_\delta/L_\delta \rightarrow 1$, (28) and (29) imply

$$\liminf_{\delta \downarrow 0} \frac{J_i^*(\delta; Q^\delta)}{L_\delta} \geq g_i \max_{j: \theta_j \neq \theta_i} \frac{1}{\text{KL}(P_i \| P_j)} = \frac{g_i}{D_i^{\text{sep}}}.$$

For the matching upper bound, run the separated pre-repair race from Section 6. Its constant decoder commits to the old label; post observations can be ignored. Its error analysis therefore depends only on the fixed pre laws. Repair success and fully charged tails still have the same fixed q , so its expected cost has coefficient g_i/D_i^{sep} . This policy is also joint-feasible for every Q^δ . Combining bounds proves the result. $\square$

For a concrete example, retain the three pre means and labels of (18) and set, for $L_\delta \geq 1$,

$$Q_i^\delta = \text{Bern}\left(\frac{1}{2} + \frac{a_i}{L_\delta}\right), \quad (a_1, a_2, a_3) = (-1/8, 0, 1/8). \quad (31)$$

The post means are always distinct and lie in $[3/8, 5/8]$. The elementary inequality $\text{KL}(\text{Bern}(u) \| \text{Bern}(v)) \leq (u - v)^2/[v(1 - v)]$ follows from $\log x \leq x - 1$. Hence all post divergences are $O(L_\delta^{-2})$, and (29) holds.

If any one of these post experiments is frozen and the confidence level is then sent independently to zero, its singleton partition admits a universal decoder and Theorem 4.2 gives bounded expected cost. Along (31), however, (30) gives the positive separated coefficient. The two limiting operations therefore have different outcomes. This is a quantified boundary of the original repair theorem, not a contradiction to it.

8 The exact cost of a declaration gate

Let V_h^G denote the optimal value with a nonempty gate $G_h(b, x)$ and with the same terminal choices as the joint controller. Define

$$A_h(b, x, a) = \mathcal{Q}_h(b, x, a) - V_h(b, x) \geq 0, \quad W_h(b, x) = V_h^G(b, x) - V_h(b, x). \quad (32)$$

The quantity A_h is a loss advantage: it is the excess cost of choosing a once and then continuing optimally without the gate.

Theorem 8.1 (Exact gate-cost characterization). *For every finite model of Section 3 and every nonempty action gate,*

$$W_0(b, x) = 0, \quad (33)$$

$$W_h(b, x) = \min_{a \in G_h(b, x)} \left\{ A_h(b, x, a) + \sum_{y, x'} q_b(y, x' | x, a) W_{h-1}(\Phi(b, x, a, y, x'), x') \right\}. \quad (34)$$

Equivalently,

$$W_T(\mu, x_0) = \min_{\pi \in \Pi_G} \mathbb{E}_\mu^\pi \left[\sum_{t=0}^{T-1} A_{T-t}(b_t, X_t, A_t) \right], \quad (35)$$

where terminal decisions are chosen optimally. Consequently:

1. $W_h(b, x) \geq \min_{a \in G_h(b, x)} A_h(b, x, a)$.
2. Every feasible gated policy gives an upper certificate by the expectation in (35).
3. $W_T(\mu, x_0) = 0$ if and only if there is a gated policy selecting a joint optimal action at every node it reaches with positive probability.

Proof. Substitute $V_{h-1}^G = V_{h-1} + W_{h-1}$ into the gated Bellman equation and subtract V_h . This proves (34). For any policy with optimal terminal decisions, conditional expectation gives

$$\mathbb{E}[\ell_t + V_{T-t-1}(b_{t+1}, X_{t+1}) - V_{T-t}(b_t, X_t) \mid \mathcal{H}_t, A_t] = A_{T-t}(b_t, X_t, A_t).$$

Sum over t . The continuation terms telescope, and the expected terminal loss equals V_0 , yielding $J_T(\pi) - V_T = \mathbb{E}^\pi \sum_t A_{T-t}$. Minimizing proves (35). All advantages are nonnegative, proving the certificates. An optimal gated finite policy tree exists. A finite sum of nonnegative node contributions is zero exactly when each positive-probability node contributes zero. This proves the last equivalence. $\square$

The theorem is an application of standard Bellman telescoping. Its use here is to make the proposed separation comparison exact. A local forbidden repair is not enough to imply a positive initial gap: an optimal policy may avoid that belief altogether. Conversely, a strict forbidden optimum at the initial belief immediately yields a positive lower certificate. The gate gap also accounts for changes in future observations caused by actions; it is not a difference between two static task costs.

8.1 An exact no-information benchmark

Assume two environments with prior $p = \mathbb{P}(I = 1)$, labels $\Theta_0(f_0) = 1$, $\Theta_1(f_0) = 0$, and a deterministic irreversible repair to f_1 with reversed labels. Old task losses are $(0, r)$ and repaired losses are $(s, 0)$, where $r, s > 0$. Repair costs $c_r > 0$, takes effect before its current task, and has no subsequent control decisions other than continuing. Neither mode gives environment information. There is no verification or fallback, and terminal penalties are symmetric, $\lambda > 0$.

Lemma 8.2 (Immediate repair or no repair). *An optimal policy either repairs now or never repairs.*

Proof. No observation changes the prior. Repairing after t old-mode rounds costs $tpr + c_r + (h - t)(1 - p)s + \lambda \min(p, 1 - p)$ for $0 \leq t < h$. Never repairing costs $hpr + \lambda \min(p, 1 - p)$. If $a = pr - (1 - p)s \leq 0$, no repair helps. If $a > 0$, the saving $(h - t)a - c_r$ is largest at $t = 0$. Randomization produces only convex combinations of these costs. $\square$

Proposition 8.3 (Threshold and exact separation gap). *In this family, for $h \geq 1$,*

$$V_h(p) = \lambda \min(p, 1 - p) + \min\{hpr, c_r + h(1 - p)s\}. \quad (36)$$

Repair is weakly optimal exactly when

$$p \geq p_{\text{int}}(h) = \frac{c_r + hs}{h(r + s)}. \quad (37)$$

With the declaration gate $p_{\text{dec}} = 1/2$,

$$W_h(p) = \mathbf{1}\{p < 1/2\} \max\{0, h[pr - (1 - p)s] - c_r\}. \quad (38)$$

The interval of strict intervention before declaration is nonempty exactly when $h(r - s) > 2c_r$.

Proof. The label reversal preserves the symmetric terminal risk, so Lemma 8.2 gives the value and threshold. If $p < 1/2$, the unchanged posterior never opens the gate and its value is the never-repair cost. If $p \geq 1/2$, the gate excludes no repair plan. Subtraction gives (38). Finally $p_{\text{int}}(h) < 1/2$ is equivalent to the stated strict inequality. $\square$

The boundary cases are explicit. A threshold above one gives an empty repair region; a threshold equal to one gives a tie only at $p = 1$. The threshold increases with c_r , and

$$p_{\text{int}}(h+1) - p_{\text{int}}(h) = -\frac{c_r}{h(h+1)(r+s)} < 0. \quad (39)$$

For $h = 10$, $r = 2$, $s = 1/2$, $c_r = 1$, $p = 3/10$, and $\lambda = 1$, $p_{\text{int}} = 6/25$, $V_h = 24/5$, and $V_h^G = 63/10$. The exact gap is $3/2$. This is a useful analytic benchmark, not by itself a deep general control result.

9 Repair regions with noisy verification

9.1 Affine pieces determine a component bound

Return to a binary environment and a fixed visible state. Before repair, any finite collection of information and control actions is allowed, with belief-independent physical availability as in Section 3. Consider one specified deterministic, irreversible repair u . Following u , there are no further intervention choices: a fixed task policy is used and a passive observation transcript Z is obtained before the terminal declaration. Let $q_i(z)$ be its finite distribution under environment i . Suppose the repaired representation has opposite adequacy labels in the two environments, terminal error costs are symmetric λ , and the expected nonterminal cost of this repair plan is affine, $a_u(p)$.

The repair-now value is then

$$R_u(p) = a_u(p) + \lambda \sum_z \min\{(1-p)q_0(z), pq_1(z)\}. \quad (40)$$

The sum uses joint prior-transcript masses. No posterior normalization is needed. Its interior breakpoints are the distinct values

$$\mathcal{B}_u = \left\{ \frac{q_0(z)}{q_0(z) + q_1(z)} : q_0(z)q_1(z) > 0 \right\}. \quad (41)$$

If there are m_u distinct breakpoints, there are at most $m_u + 1$ affine pieces. Transcripts possible in only one environment contribute zero decision risk.

Theorem 9.1 (A repair-region component bound). *Under the preceding assumptions, the set of priors where repair u is weakly optimal is the union of at most $m_u + 1$ closed intervals, allowing empty intervals and singletons. On every affine piece of R_u , its intersection with the repair region is an interval. If the post-repair transcript is uninformative, the repair region has at most two connected components.*

Proof. Let $C_u(p)$ be the optimal value among all policy trees whose first action is not u . It is a minimum of finitely many affine functions of p by Proposition 3.2; hence it is concave and continuous. The repair region is $\{p : R_u(p) \leq C_u(p)\}$. On any closed piece where R_u is affine, $C_u - R_u$ is concave, and its nonnegative superlevel set is convex. A closed convex subset of an interval is an interval, a singleton, or empty. Taking the union over at most $m_u + 1$ pieces proves the bound. If $q_0 = q_1$, the decision risk is $\lambda \min(p, 1-p)$ with the single breakpoint $1/2$. $\square$

The result applies regardless of the complexity of the optimal *pre-repair* continuation. It needs no TP2 assumption on verification. It does need the specified passive continuation after repair; allowing subsequent controlled repairs replaces (40) by a more complicated piecewise-linear continuation value and does not preserve the two-component bound automatically. The bound is about components of one action’s optimality region, not the piece count of the entire optimal policy.

Corollary 9.2 (Independent binary observations after repair). *If repair is followed by n independent Bernoulli observations with distinct full-support success probabilities in the two environments, then $m_u \leq n + 1$ and the repair region has at most $n + 2$ connected components.*

Proof. The likelihood ratio of a transcript depends only on its number of successes, which takes $n + 1$ values. Apply Theorem 9.1. $\square$

This is an upper bound, not a claim that every number of components is attainable for every channel. More informative observations improve optimal value under the conditions in Section 10, but component counts and region inclusion need not be monotone.

9.2 A sharp two-region example with a TP2 channel

Consider one remaining round, symmetric terminal cost $\lambda = 2$, old task losses $(0, 2)$, and repaired task losses $(1/5, 0)$. Repair is deterministic, irreversible, costs $1/10$, and yields no environment information. Verification costs $1/2$, replaces that round’s ordinary task (zero additional task loss), leaves the representation unchanged, and observes a binary signal with

$$P(Y = 1 \mid I = 0) = \frac{1}{5}, \quad P(Y = 1 \mid I = 1) = \frac{4}{5}. \quad (42)$$

Both observation distributions have full support. The channel matrix with ordered states and signals has determinant $3/5 > 0$ and is TP2. The three action values are

$$K(p) = 2p + 2 \min(p, 1 - p), \quad (43)$$

$$R(p) = \frac{1}{10} + \frac{1 - p}{5} + 2 \min(p, 1 - p), \quad (44)$$

$$B(p) = \frac{1}{2} + 2 \min\left\{p, \frac{1}{5}, 1 - p\right\}. \quad (45)$$

The verification formula follows by summing the smaller joint mass for each signal. All three actions share the same terminal loss contract.

Theorem 9.3 (Disconnected repair and strict early intervention). *In this full-support noisy-verification instance, the weak repair region is*

$$\mathcal{R} = \left[\frac{3}{22}, \frac{1}{3}\right] \cup \left[\frac{7}{11}, 1\right]. \quad (46)$$

The keep region is $[0, 3/22]$ and the verify region is $[1/3, 7/11]$, with shared endpoints denoting ties. In particular, TP2 verification alone does not guarantee a single repair interval. For every $p \in (3/22, 1/3)$, repair is the unique optimal action while the immediate declaration benchmark prefers adequate.

Proof. For $p \leq 1/2$, $R(p) = 3/10 + 9p/5$; for $p \geq 1/2$, $R(p) = 23/10 - 11p/5$. Comparing R and K gives $p \geq 3/22$. The verification value equals $1/2 + 2p$ on $[0, 1/5]$, $9/10$ on $[1/5, 4/5]$, and $5/2 - 2p$ on $[4/5, 1]$. On the middle interval, $R \leq B$ holds for $p \leq 1/3$ or $p \geq 7/11$. On the two exterior intervals, $R < B$. Combining the inequalities gives (46); comparing K and B on the remaining pieces gives the other regions. Strict inequalities hold in the stated interiors. Finally $p < 1/2 = p_{\text{dec}}$ throughout $(3/22, 1/3)$. $\square$

At $p = 1/4$, the three costs are $K = 1$, $B = 9/10$, and $R = 3/4$. The joint controller repairs, while the declaration gate permits only keep or verify and chooses verify. Its exact extra cost is $3/20$. At $p = 1/2$, verification is preferable; at sufficiently high p , repair is again preferable. A single threshold therefore misses an optimal repair region even though the pairwise repair–keep comparison has a threshold.

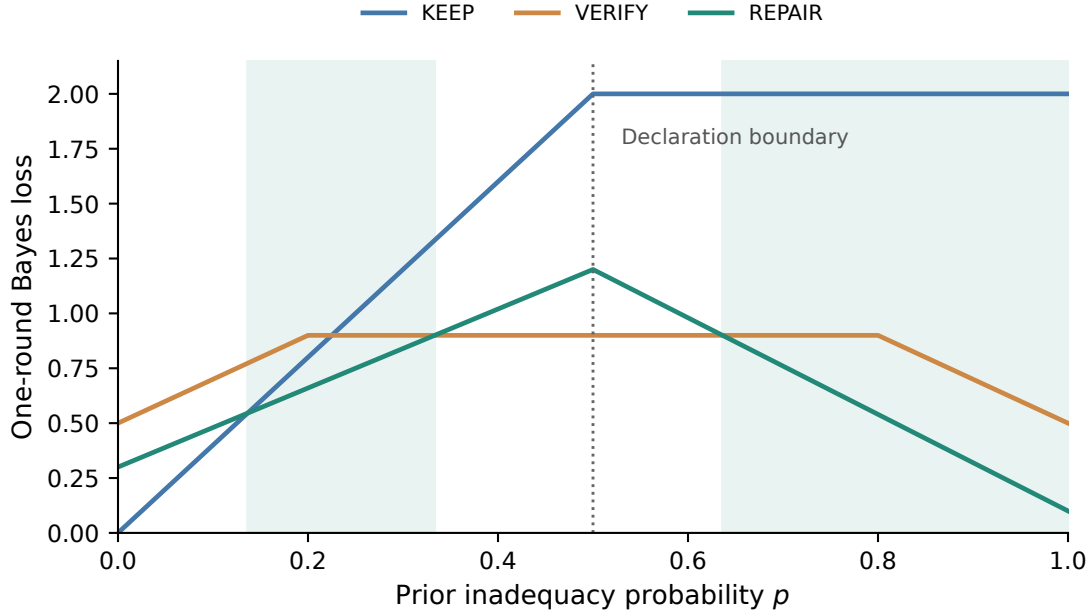


Figure 1: Exact one-round action values. Shading marks the two repair intervals. The dotted line is the declaration boundary, not an action constraint for the joint controller. The lower repair interval lies entirely below that boundary.

9.3 Robustness beyond one parameter point

Proposition 9.4 (Finite-horizon perturbation certificate). *Consider two models with the same finite environment, states, actions, observation alphabet, and terminal decision sets. Suppose corresponding kernel rows differ by at most ϵ_Q in total variation, stage costs differ by at most ϵ_ℓ , and terminal costs differ by at most ϵ_L . Both stage and terminal costs are bounded by $L_{\max}$ and L_B , respectively. For any fixed horizon- h policy tree, and hence for an optimal value or a root-action-constrained optimal value, the absolute value difference is at most*

$$\mathcal{E}_h = h\epsilon_\ell + \epsilon_L + \epsilon_Q \left\{ L_{\max} \frac{h(h+1)}{2} + hL_B \right\}. \quad (47)$$

A unique action with gap $g > 2\mathcal{E}_h$ remains uniquely optimal.

Proof. Condition on the same environment and couple corresponding observable events maximally while histories agree. The probability that histories differ by the end of round t is at most $t\epsilon_Q$. The round- t loss difference is bounded by $\epsilon_\ell + tL_{\max}\epsilon_Q$ for $t = 1, \dots, h$; the terminal difference is at most $\epsilon_L + hL_B\epsilon_Q$. Summation gives (47). The bound is uniform over policy trees. Taking minima over the same feasible set preserves it, including when the first action is fixed. Two action values can move toward one another by at most $2\mathcal{E}_h$. $\square$

At $p = 1/4$, the strict repair margin is $3/20$. Continuity or the explicit bound therefore gives a neighborhood of costs and full-support channel probabilities where early repair remains uniquely optimal. The prior is also separated from the declaration boundary. Thus the phenomenon

holds on a nonempty open parameter set, not only at a rational point. This perturbation statement keeps physical action sets fixed; it does not assert continuity across arbitrary changes in belief-dependent gates.

10 What information can eliminate

10.1 Blackwell improvement for a pure information action

Proposition 10.1 (Standard information monotonicity). *Replace a pure verification channel P by $\tilde{P}$ on the same hidden environment, where $P = \tilde{P}G$ for an environment-independent stochastic garbling matrix G . Suppose verification has the same environment-wise cost in both models, independent of its signal, and leaves the visible state unchanged (or makes the same deterministic budget transition); all other primitives are unchanged. Then the unrestricted optimal Bayes loss cannot increase, at any finite horizon or initial belief.*

Proof. After each fine-channel observation, a controller can sample the garbling and run any coarse-channel policy on the resulting fictitious coarse history. Its joint law of environments, coarse observations, actions, states, and losses matches that in the coarse model. The fine model admits this randomized policy, and a deterministic optimum is no worse. Taking the infimum over coarse policies proves the result. $\square$

This is the usual experiment-comparison argument (Blackwell, 1953). It compares total optimal value, not a static ranking of arbitrary interventions. Changing the representation, state transition, or verification cost violates the comparison’s assumptions. Also, a gate based on the full posterior can change which actions are permitted after refinement, so the same simulation argument does not automatically prove monotonicity of the gated optimum.

10.2 An informative gate need not have a linear cost

Augment the binary family of Proposition 8.3 with a repeatable verification action. Each verification consumes a round, leaves the old representation in place, incurs its old task loss plus $c_v > 0$, and yields an independent observation from P_i under environment i . Assume a finite alphabet, distinct full-support P_0, P_1 , deterministic repair, and no post-repair information. There is no fallback or deferral. Keep and repair retain their earlier meanings and costs, including the symmetric terminal cost. All parameters are fixed as T increases.

For $0 < p < 1$, define

$$\rho = \sum_y \sqrt{P_0(y)P_1(y)} < 1, \quad \beta = -\log \rho > 0. \quad (48)$$

Consider the following feasible gated policy: verify exactly $n \leq T - 1$ times, estimate the environment by the posterior MAP rule, repair if it estimates $I = 1$, otherwise keep, and then execute tasks to the horizon. Report the adequacy label implied by the inferred environment and the final representation. On a posterior tie, choose $I = 1$, which is gate-feasible.

Theorem 10.2 (Finite bound and logarithmic gate-cost upper bound). *The preceding policy satisfies*

$$J_T(\pi_n) \leq n(pr + c_v) + c_r + [(T - n) \max(r, s) + \lambda] \sqrt{p(1 - p)} e^{-\beta n}. \quad (49)$$

For all sufficiently large T , choose $n_T = \lceil 2 \log T / \beta \rceil \leq T - 1$. Then

$$0 \leq V_T^G(p) - V_T(p) \leq J_T(\pi_{n_T}) = O(\log T). \quad (50)$$

Consequently, a lower bound $cT - C$ with fixed $c > 0$ cannot hold for this informative family for all sufficiently large horizons.

Proof. Let z be the length- n verification transcript. The Bayes MAP error is

$$\begin{aligned}\epsilon_n &= \sum_z \min\{(1-p)P_0^n(z), pP_1^n(z)\} \\ &\leq \sqrt{p(1-p)} \sum_z \sqrt{P_0^n(z)P_1^n(z)} = \sqrt{p(1-p)}\rho^n,\end{aligned}$$

using $\min(a, b) \leq \sqrt{ab}$ and independence. The verification phase has expected cost $n(pr + c_v)$. There is at most one repair costing c_r . If the MAP decision is correct, the selected representation has zero remaining task loss; if it is wrong, the loss per remaining round is at most $\max(r, s)$. The terminal error probability is ϵ_n , because the chosen representation's label agrees with the declaration exactly when the environment estimate is correct. This proves (49). A repair occurs only at a posterior at least $1/2$, so the policy is gate-feasible. With n_T the exponential term is at most T^{-2} , and the first term grows at most logarithmically. Finally $V_T \geq 0$ and $V_T^G \leq J_T(\pi_{n_T})$ give (50). $\square$

The theorem is an upper bound, not a matching characterization of the gate gap. The gap may be bounded or zero in other instances. The bound also does not extend to a fixed verification budget, unidentifiable channels, costs growing with T , or repairs with nonzero residual loss in their intended environments without further analysis.

For numerical illustration, we evaluate fixed-prefix policies exactly using binomial counts with $p = 3/10$, $P_0(Y = 1) = 1/5$, $P_1(Y = 1) = 4/5$, $r = 2$, $s = 1/2$, $c_r = 1$, $c_v = 1/4$, and $\lambda = 1$. For each plotted horizon, we minimize over $0 \leq n \leq \min(100, T - 1)$. This is the best *tested prefix*, not the unrestricted gated optimum. The environment oracle has loss $pc_r = 3/10$ for the reported horizons, and $J_T(\pi_n) - 3/10$ is a valid upper bound on the gate gap.

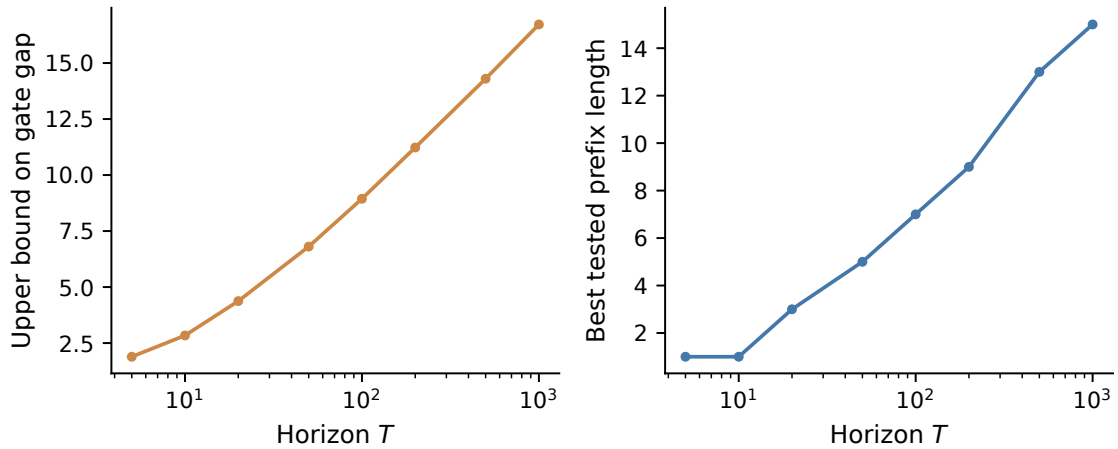


Figure 2: A constructive upper bound when verification is repeatable and identifiable. Left: tested fixed-prefix policy cost minus the oracle lower bound. Right: its selected prefix length. These are not measurements of the exact gated optimum at the large horizons.

10.3 Why the two boundaries coexist

Equation (38) concerns a controller that cannot acquire new environment evidence. Equation (50) concerns one that can repeatedly buy informative evidence. The former does not imply a universal linear separation principle, and the latter does not remove the finite-horizon counterexample of Theorem 9.3. A short deadline, the task cost of verification, and what repair makes observable can all influence the optimal sequence. They must be compared under one explicit action and loss contract.

11 Experiments and independent computation

All experiments are specified finite models. Bayes values and posterior updates use Python rational arithmetic. Certification likelihood scores and count-lattice probabilities use double precision with stable binomial tails. No Monte Carlo experiment, participant study, or learned kernel is reported.

11.1 Old-label certification and finite accounting

The certification configuration uses (18), $q = 0.8$, $\kappa_i = 0.3$, $\delta = 0.05$, $H = 2400$, $N = 2000$, and $R = 24$. These finite budgets are explicit; they are not obtained by substituting $\delta = 0.05$ into the asymptotic sufficient schedule H_δ . Both policies use $\beta = \log 480$. The joint race permits every decoder and the separated race permits only constant ones. The factorized implementation uses the same first-decoder tie rule as exhaustive search.

The evaluator propagates probability mass on Bernoulli count states (n, u) . It checks eligibility also at $n = N$. An unresolved history incurs $g_i H$ and uses the pre-history maximum-likelihood old label. A stop at n incurs conditional expected cost

$$g_i n + \kappa_i \frac{1 - (1 - q)^R}{q} + (1 - q)^R g_i (H - n - R).$$

All-failed separated histories retain their committed answer. Joint histories use a pre-history fallback. On each possible success time the remaining post count is evaluated separately; replacing repair duration by its expectation would not give the same terminal error.

Policy	Env.	Cost	$\mathbb{E}N_P$	Error	Timeout
Joint	E_1	13.509374	13.134374	4.15e-18	8.71e-152
Joint	E_2	28.562326	28.187326	1.54e-03	2.45e-73
Joint	E_3	21.486990	21.111990	1.67e-03	1.82e-73
Certify first	E_1	306.420695	306.042190	1.91e-03	8.77e-06
Certify first	E_2	28.648038	28.273038	1.23e-05	3.44e-73
Certify first	E_3	309.903293	309.524760	3.57e-03	8.84e-06

Table 2: Finite count-lattice evaluation of old-label certification. Displayed errors and timeout masses are floating-point values. Every total cost includes failed attempts and unrepaired tails.

Every directly evaluated error is below 0.05, and every conservative bound from Section 7 with the nearest-mean error bound is below 0.01251. In E_1 the costs are 13.509374 and 306.420695, giving a finite ratio 22.682079 rather than the asymptotic ratio 25.026980. The improvement varies across environments. The E_1 joint unresolved mass is approximately 8.71×10^{-152} . Numerical total-mass residuals around 10^{-16} are reported separately and are not substituted for this tail probability.

11.2 A finite-deadline boundary of free post information

For (31), we evaluate the proved lower bound (28) at $L_\delta = 2, 4, 8, 16, 32, 64, 128, 256$, using $H = L_\delta^2$. These curves are analytic lower-bound evaluations; no purported exact large-horizon certification optimizer is run. In E_1 the normalized bound approaches the separated coefficient 48.9931965, as proved by Theorem 7.2. Every frozen post experiment has a universal decoder, but the varying family has this positive leading coefficient.

11.3 Current-label control in the original three-environment fixture

This separate Bayes fixture has environments adequate, repairable, and unrepairable, prior $(0.6, 0.25, 0.15)$, horizon $T = 6$, and symmetric terminal penalty $\lambda = 0.5$. Old and new labels are

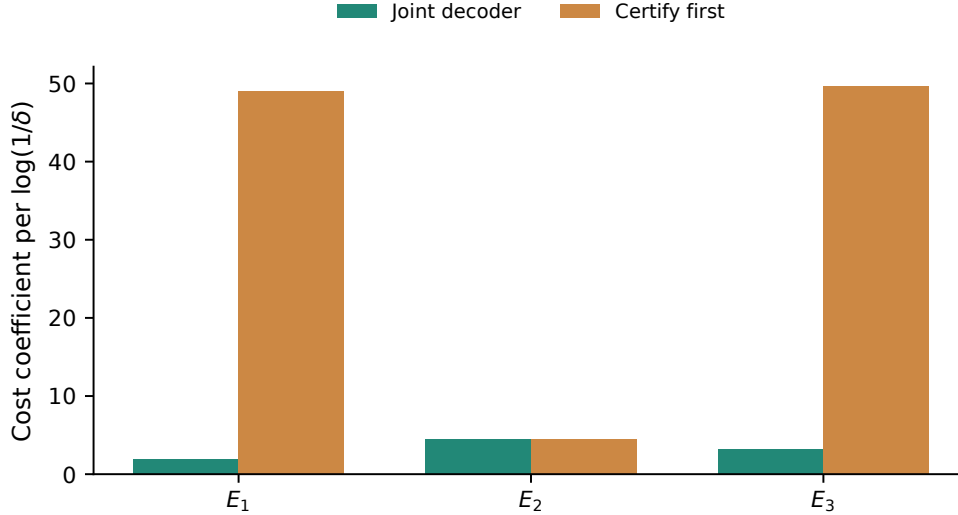


Figure 3: Fixed-instance asymptotic coefficients in the three-environment example. Bars compare policy classes under the same old-label contract.

$(1, 0, 0)$ and $(1, 1, 0)$; their ordinary task costs are $(0, 0.5, 0.5)$ and $(0, 0, 0.5)$. Every action takes one slot. In old mode, KEEP has no information, VERIFY adds 0.08 to the old task loss and yields Bernoulli means $(0.2, 0.8, 0.8)$, and REPAIR adds 0.2 to old task loss and succeeds with probability 0.8 in every environment. Failure leaves the old mode available for another attempt. The repair affects the *next* task. New-mode tasks yield Bernoulli means $(0.1, 0.1, 0.9)$. Fallback costs 0.12 per round, gives no information, and preserves whether the installed representation is old or new. There are four visible modes.

The original gate restricts REPAIR only, at posterior threshold 0.95; fallback remains available. This is neither the old-label commitment class nor exactly the two-action gate definition used for the binary theorems. Both its action set and its threshold are recorded in the model. The joint cost

Policy	Bayes cost	First action	e_1	e_2	e_3
joint	0.73146152	repair	0.036824	0.036824	0.017384
gate_095	0.83414400	verify	0.232000	0.072000	0.072000
no_repair	0.83414400	verify	0.232000	0.072000	0.072000

Table 3: Exact current-label Bayes fixture, rounded for display. The e_i columns are conditional terminal errors under the same implemented policy; no uniform δ constraint is imposed.

is exactly $9143269/12500000 = 0.73146152$; the two restricted costs are $26067/31250 = 0.834144$. Initial actions are repair and verify, respectively. Conditional errors agree with the independently reconstructed original values.

A threshold sweep also yields 0.834144 at threshold 0.5 for this same repair-only gate. Thus the default gap is not solely an artifact of imposing 0.95. The sweep evaluates explicit alternative restrictions and does not turn a posterior gate into a frequentist confidence guarantee.

One-at-a-time sensitivity varies repair fee over 0, 0.1, 0.2, 0.4, 0.8, success probability over 0, 0.2, 0.5, 0.8, 1, verification accuracy over 0.5, 0.65, 0.8, 0.95, 1, and horizon over 1, 2, 4, 6, 8. All values and first actions are in Appendix K. The maximum difference from the supplied source archive’s cost and error columns is approximately 2.3×10^{-13} . Timeout residuals are treated separately as described above.

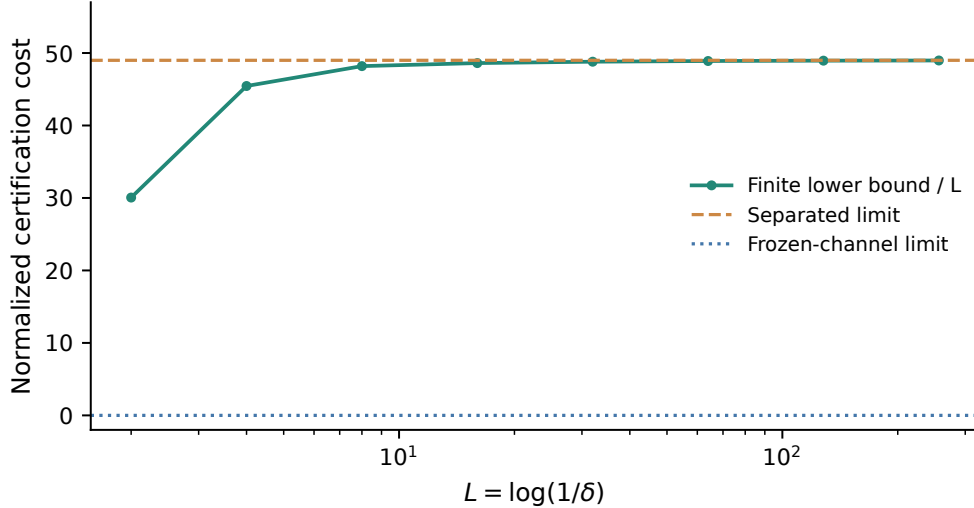


Figure 4: For E_1 , finite information lower bound divided by L_δ along the shrinking-channel family, compared with its separated limit. The zero line is the coefficient obtained when any one of the distinct post experiments is frozen before sending the confidence level to zero.

11.4 Binary families and repair-region geometry

Nine binary configurations retain the finite-horizon development: F0 fixes the representation; F1 is the no-information benchmark; F2 uses perfect verification; F3 uses full-support noisy verification; F4 allows fallback and deferral; F5 has one repair attempt with success probabilities $(3/4, 1/2)$; F6 adds passive post information; F7 is unidentifiable; and F8 is the exact one-round two-region example. Complete rational parameters are stored in the configurations. Unlike the three-environment fixture, these repairs affect the current task. For F0–F7 verification retains old task loss in addition to its price. For F8 and the repair-region sweep it replaces the round’s ordinary task. No comparison changes this convention between policies.

The joint and gate policies optimize their respective menus exactly. Stage-myopic minimizes immediate cost; greedy-VOI minimizes immediate cost plus the one-action terminal Bayes risk. Verify-then-act verifies once initially and subsequently disables verification. Never-repair retains other actions; always-repair forces the first repair and then optimizes. All use the same optimal terminal declaration. The environment oracle knows the true index only to provide a lower bound. F1 gives the exact gap $3/2$ and F8 the noisy gap $3/20$. F4 has equal joint and gate

Instance	Oracle	Joint	Gate	Myopic	Gate gap
F0	2.4000	3.0000	3.0000	3.0000	0.0000
F1	0.3000	4.8000	6.3000	6.3000	1.5000
F2	0.0300	1.1200	1.1300	1.1200	0.0100
F3	0.0300	1.4000	2.1300	1.4000	0.7300
F4	0.3000	2.6400	2.6400	3.6000	0.0000
F5	0.7200	2.3075	2.6520	2.3700	0.3445
F6	0.0300	0.9466	1.9290	0.9466	0.9825
F7	0.7800	1.9750	3.6000	1.9750	1.6250
F8	0.0250	0.7500	0.9000	0.7500	0.1500

Table 4: Binary Bayes benchmarks. Exact fractions and all seven policy columns are provided in the CSV. A positive gate gap is not universal.

values. Comparing F3 with F6, passive information reduces joint loss from 1.4 to 0.94656 and gate loss from 2.13 to 1.92904, while their gap increases. Better information therefore need not

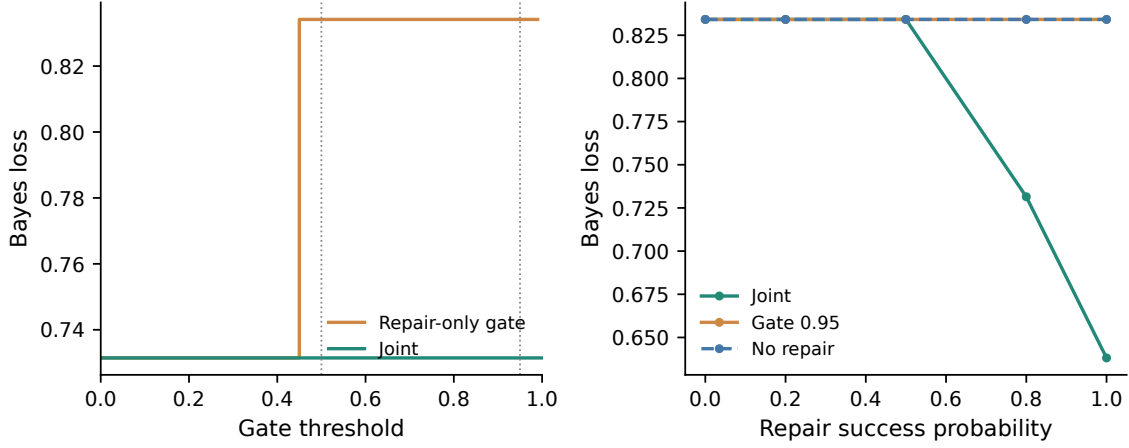


Figure 5: Current-label fixture. Left: repair-only gate as its threshold varies, with unrestricted value as a reference. Right: values as repair success varies at the original 0.95 gate. Policies share all other primitives within each comparison.

reduce the difference between these two policy classes.

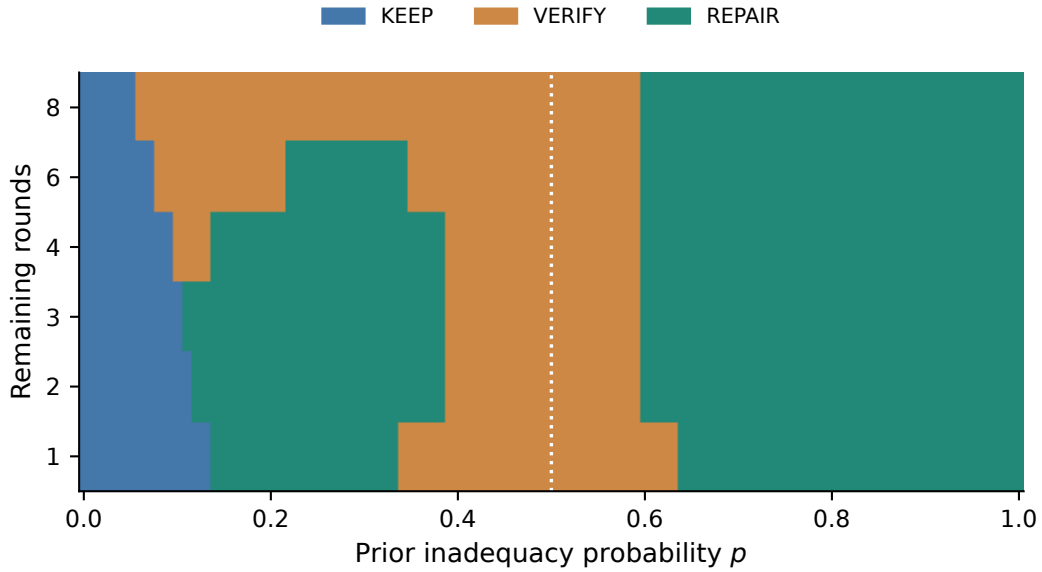


Figure 6: Exact actions on 101 rational priors at six horizons. Interpolation between sampled priors is illustrative; only the one-round boundaries in Theorem 9.3 have the stated analytic endpoints.

The long-horizon fixed-prefix illustration in Figure 2 tests $0 \leq n \leq \min(100, T - 1)$. At $T = 1000$ it selects $n = 15$ and has loss approximately 17.02328. Subtracting the oracle 0.3 gives a valid gate-gap upper bound 16.72328, not the optimal gated value.

11.5 Independent checks and computational limits

The policy-tree enumerator never updates a posterior or invokes the Bellman solver. It enumerates environment-wise loss vectors for all deterministic trees, retaining duplicates, and then minimizes their prior inner products. All 20 binary prior–horizon comparisons and all 12 three-environment comparisons have exactly zero rational difference. The largest enumerated tree family has 61,872

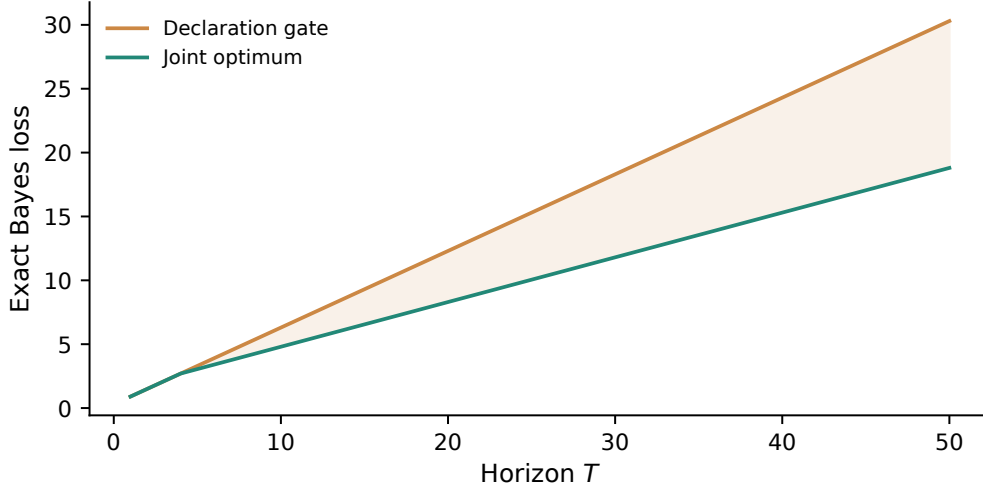


Figure 7: The exact no-information gap is $\max(0, T/4 - 1)$ at the F1 parameters. All displayed values agree with the rational solver.

policies.

The 32-test suite checks exact thresholds and endpoints, factorized versus enumerated decoder coefficients and eligibility, equality-at-threshold ties, global decoder tie breaking, the two finite protocol implementations, visible success information, aggregation, multi-slot action completion, fallback labels, asymmetric errors, abstention, serialization, gate menus, and a 10^{-12} -probability branch with 10^{12} loss. Tests of the new structural formulas supplement their proofs; they do not establish mathematical priority or formal verification.

For N_b cached belief nodes, at most A actions and O joint observable outcomes, Bellman computation uses $O(N_b A O K)$ rational operations, excluding integer bit complexity. The number of nodes can grow exponentially in the horizon. The supplied binary timing family grows from five cached nodes at one round to 313 at twelve rounds; this is an instance-specific count. An online factorized decoder check is polynomial in K , but a Bernoulli count-lattice evaluator handles $O(N^2)$ histories, each with its own eligibility and post-tail calculation. These are separate complexity claims.

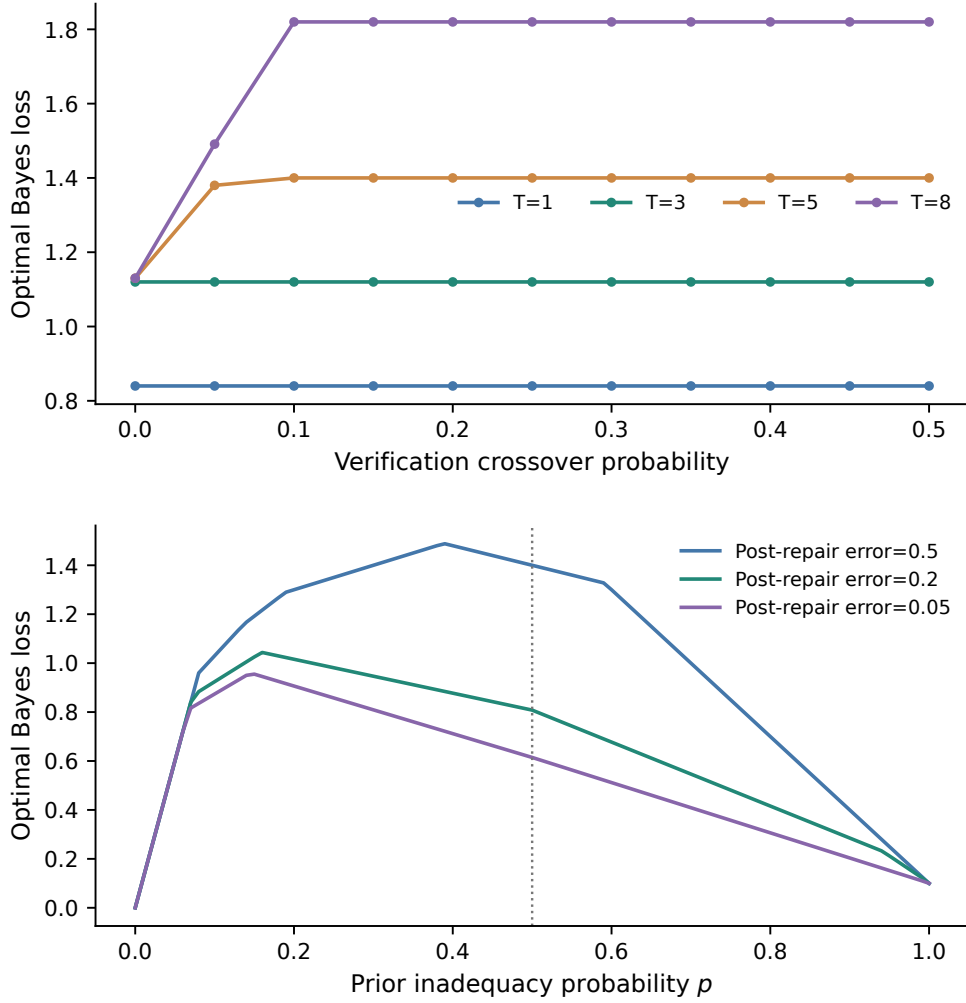


Figure 8: Top: unrestricted value under a pure verification channel comparison at fixed costs. Bottom: values for three passive post-repair channels. These are finite value calculations, not general monotonicity claims for the gate gap.

12 Discussion and limitations

The old-label result makes repair useful as an information operation even when the eventual answer remains about the obsolete representation. The useful pre-repair answer is a decoder for the information available later. The classwise bottleneck and surviving-set test make this answer structure explicit and computationally manageable for a single online trajectory.

The fixed-instance coefficient depends on exact equality of post laws. The finite-deadline bound supplies the missing quantitative qualification: a nominally distinguishing channel may contribute too little information within the available time. The shrinking-channel theorem demonstrates that the leading advantage can disappear without any two post laws being exactly equal at a finite confidence level. It also explains why finite experiments should report actual post budgets rather than only the partition.

The current-label Bayes results answer a different operational question. A declaration gate can forbid a uniquely optimal repair, and the cost of the restriction depends on where the policy actually goes. A pairwise repair-keep threshold need not remain a threshold when verification is available. When repair has a fixed passive continuation, its terminal risk pieces bound the number of disconnected regions regardless of the pre-repair policy’s complexity.

The following limits determine what can be inferred.

1. Models, costs, and candidate representations are known and finite. The paper neither estimates unknown kernels nor learns a new representation family. Exact risk-table realization alone imposes no additional control structure.
2. The fixed-confidence coefficient assumes one pre channel, no return to pre observations after entering repair, fixed environment-independent success probability, and zero post cost. Positive post costs or multiple controlled channels call for a new allocation analysis.
3. Environment-dependent success changes the information in repair outcomes, so the random-decoder proof cannot be reused unchanged. For fixed $q_i > 0$, however, the expected attempts until success can still be bounded by $1/q_i$. Informative outcomes alone do not prove that repair cost becomes leading order.
4. The two-component Bayes repair bound requires a binary environment, one specified deterministic irreversible repair, passive continuation, and symmetric terminal error cost. The generic solver supports some cases, such as stochastic repair, outside this theorem.
5. The $1/2$ declaration gate uses symmetric Bayes errors. An externally imposed 0.95 posterior gate is a different restriction. Neither is the uniform fixed-confidence commitment requirement.
6. The logarithmic Bayes gate bound is an upper bound in a specified zero-residual-loss family. No matching logarithmic lower bound or optimal constant is asserted. The finite information lower bound is also not a general solution of paid two-phase sensing.
7. Computational examples are small, synthetic, and deterministic. Count-lattice certification sums use floating point, while the Bayes solver uses exact rational values. Neither confirms a general theorem by experiment nor establishes large-scale deployment performance.

Further questions. A natural next theory problem charges observations on both sides of repair, or chooses among several repairs with different post partitions and residual task risks. Another imposes a meaningful constraint on representations, such as a fixed feature or memory budget, and asks which additional control structure follows. These questions are more specific than assuming that representation change must create a fundamentally non-Markovian process. A belief state remains sufficient for the finite known model considered here.

13 Conclusion

Repair can change which evidence must be collected before acting and how later evidence is interpreted. In an ordered fixed-confidence experiment this gives a decoder coefficient, a factorized computation, and a polynomial likelihood-ratio policy. A finite-deadline lower bound identifies a regime in which weakening post observations removes the coefficient advantage. For current-label finite-horizon control, declaration gates have an exact cost characterization, and a passive repair continuation bounds the topology of repair regions. Explicit noisy examples and a reproducible implementation make the distinction between certification and intervention checkable.

Reproducibility and computational assistance. The accompanying release contains the complete manuscript and proofs, LaTeX, bibliography, exact Bayes solver, independent policy-tree enumerator, two decoder implementations, frozen configurations, numerical outputs, and figure-generation commands. The provided source archive for the earlier manuscript contained LaTeX and generated outputs; its original Python repository was not supplied. The integrated release uses the independently reconstructed evaluators described here and does not represent the unavailable repository as audited. Mathematical development, drafting, and programming used AI assistance. This is a research draft, without a claim of formal proof verification, peer-review

acceptance, or confirmed priority over all related literature. No participant data or private deployment logs are used.

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A Notation and theorem assumptions

Symbol	Meaning
$I, \mathcal{E}, \mu$	Fixed hidden environment, its finite set, and prior
F_t, M_t, B_t	Representation, visible policy mode, and repair budget
$X_t, \mathcal{H}_t$	Visible control state and complete observable history
$\Theta_i(f), \Delta_i(f)$	Reference-task adequacy and excess risk
$K_i(y, x' \mid x, a)$	Joint observation–successor kernel
b_t, Φ	Posterior on environments and its Bayes update
T, h	Deployment horizon and remaining rounds
V_h, Q_h	Unrestricted optimal loss and action value
G_h, V_h^G, W_h	Gate, gated loss, and their difference
A_h	Joint-optimal Bellman loss advantage
$p_{\text{dec}}, p_{\text{int}}$	Declaration and intervention boundaries when defined
$q_i(z), m_u$	Passive post-repair transcript law and its distinct risk breakpoints
ρ, β	Hellinger affinity and its negative logarithm for verification

Result	Assumptions beyond the finite model
Decoder coefficient	Assumption 4.1; fixed instance and fully charged tails
Factorized race	Binary labels on a disjoint post-law partition
Shrinking-channel limit	Fixed pre laws and costs; $H_\delta \text{KL}(Q_i^\delta \ Q_j^\delta) = o(L_\delta)$ for opposite labels
Gate-cost identity	Nonempty gate, identical terminal decisions and loss
No-information threshold	Binary labels reversed by deterministic repair; no new information; symmetric terminal costs; only keep/repair
Repair-region bound	Binary environment; a fixed irreversible repair; passive continuation; affine nonterminal cost; opposite repaired labels; symmetric terminal costs
Two-region example	Explicit one-round parameters in Section 9.2
Perturbation bound	Common finite spaces and physical action sets; uniform kernel/cost perturbation bounds
Blackwell comparison	Pure information action, unchanged costs and state transition, environment-independent garbling
Logarithmic gate bound	Repeatable independent full-support identifiable verification; old-task cost during verification; deterministic repair; zero task loss in the correctly selected mode; fixed parameters

Each result has a proof in the main text. The tables specify their scope and should not be read as extending one theorem to another experiment family.

B A complete noisy posterior calculation

At the counterexample prior $p = 1/4$, verification yields

$$\begin{aligned}
 \mathbb{P}(Y = 1) &= \frac{3}{4} \frac{1}{5} + \frac{1}{4} \frac{4}{5} = \frac{7}{20}, & \mathbb{P}(I = 1 \mid Y = 1) &= \frac{4}{7}, \\
 \mathbb{P}(Y = 0) &= \frac{13}{20}, & \mathbb{P}(I = 1 \mid Y = 0) &= \frac{1}{13}.
 \end{aligned}$$

The representation remains f_0 , whose inadequate environment is $I = 1$. After $Y = 1$, the optimal declaration is inadequate, with conditional loss $2(3/7) = 6/7$. After $Y = 0$, it is adequate, with conditional loss $2/13$. The expected terminal loss is

$$\frac{7}{20} \frac{6}{7} + \frac{13}{20} \frac{2}{13} = \frac{2}{5}.$$

Adding the verification cost $1/2$ gives $9/10$. Repair costs $1/10$, changes the per-round expected task loss to $(3/4)(1/5) = 3/20$, and changes the label map. Its symmetric terminal risk is still $2 \min(1/4, 3/4) = 1/2$, so its total is $3/4$. All task and terminal terms are accounted for.

C Asymmetric declarations and abstention

For old labels $(1, 0)$, define

$$g_0(p) = \min\{\lambda_{\text{miss}}p, \lambda_{\text{FA}}(1 - p)\}.$$

After a label-reversing repair, the corresponding risk is

$$g_1(p) = \min\{\lambda_{\text{miss}}(1 - p), \lambda_{\text{FA}}p\}.$$

With an abstention cost, include $\lambda_{\perp}$ as a third option in each minimum. In the no-information family the two macro values are $hpr + g_0(p)$ and $c_r + h(1 - p)s + g_1(p)$. They need not share a terminal term. Thus the symmetric closed-form threshold cannot be carried over by merely substituting a new p_{dec} . One can instead partition $[0, 1]$ at the breakpoints of g_0, g_1 and compare the resulting affine expressions on each piece. The solver implements the full terminal loss without this simplification.

D The scalar adequacy probability can lose necessary information

Consider three environments, two representations, and one remaining round. Old task losses are $(0, 2, 2)$; repaired task losses are $(1/5, 0, 3)$. The labels are determined by whether these risks vanish. Repair costs $1/10$, there are no observations, and both terminal error penalties equal $1/10$. At beliefs $b = (1/2, 1/2, 0)$ and $b' = (1/2, 0, 1/2)$, the old inadequacy probability is the same, namely $1/2$.

Under b , keep costs $1 + 1/20 = 21/20$ and repair costs $1/10 + 1/10 + 1/20 = 1/4$. Under b' , keep still costs $21/20$, whereas repair costs $1/10 + 1/10 + 3/2 = 17/10$; its current label is certainly inadequate, so its terminal risk is zero. Thus the optimal action is repair at b and keep at b' . Proposition 3.1 realizes both loss tables by finite representations with environment-independent visible outputs. The scalar probability cannot be the general sufficient state.

E Independent policy-tree recurrence

Let $\Gamma_0(x) = \{(L_i(x, d))_{i=1}^K : d \in \mathcal{D}(x)\}$, retaining distinct policy labels even when their vectors coincide. For an action a and one selected continuation vector $\alpha^{y, x'} \in \Gamma_{h-1}(x')$ at each observable event, define

$$\alpha_i = \sum_{y, x'} \mathbb{K}_i(y, x' \mid x, a) \left[\ell_i(x, a, y, x') + \alpha_i^{y, x'} \right].$$

Enumerating every action and every such selection forms $\Gamma_h(x)$. By conditioning on the first event, this lists all deterministic policy-tree loss vectors. Therefore

$$V_h(b, x) = \min_{\alpha \in \Gamma_h(x)} b^\top \alpha.$$

This recurrence provides both the standard geometry argument and an independent finite implementation that never computes a posterior. It is used only for the unrestricted model; a belief-dependent gate would require checking tree feasibility at each reached posterior.

F Exact prefix-policy calculation

For a BSC with crossover e , write

$$q_0(k) = \binom{n}{k} e^k (1-e)^{n-k}, \quad q_1(k) = \binom{n}{k} (1-e)^k e^{n-k}.$$

Let $\mathcal{K}_n = \{k : pq_1(k) \geq (1-p)q_0(k)\}$, $e_0 = \sum_{k \in \mathcal{K}_n} q_0(k)$, and $e_1 = \sum_{k \notin \mathcal{K}_n} q_1(k)$. For $n \leq T-1$, the fixed-prefix policy used in Figure 2 has exact cost

$$\begin{aligned} J_T(\pi_n) = & n(pr + c_v) + (1-p)e_0[c_r + (T-n)s] \\ & + p(1-e_1)c_r + pe_1(T-n)r + \lambda[(1-p)e_0 + pe_1]. \end{aligned}$$

Every term is evaluated with rational arithmetic before decimal export. This is a direct evaluation of a feasible policy; the large-horizon experiment does not run a purported exact POMDP solver at $T = 1000$.

G Reproduction protocol and artifact map

From the project root, `python3 scripts/reproduce.py` runs the test suite, regenerates frozen computations, creates vector figures, and compiles the manuscript using `latexmk`. Without a TeX installation, use the explicit `-skip-pdf` option. The exact solver itself needs only Python's standard library. Figure generation requires NumPy and Matplotlib. The manuscript uses ordinary LaTeX packages and BibTeX.

The source is organized into model validation, belief updates, exact Bellman control, exhaustive enumeration, baseline evaluation, analytic formulas, and a command-line interface. Configuration files specify rational parameters. Result files include every table column, exact fractions, diagnostics, source and model hashes, finite enumeration checks, timing measurements, and a complete optimal policy DAG for the failed-repair instance. The generic model JSON stores full joint kernels and terminal costs.

The tests verify implementation properties independently where possible. Numerical sweeps verify finite instances, not general claims. The package does not contain a proof-assistant formalization. General multiple-repair topology, a tight noisy gate-gap asymptotic constant, and open-world representation discovery remain outside the proved scope.

H Additional code validation for the three-environment example

The test suite constructs the full explicit kernel for the preceding three-environment example and checks both optimal actions without compressing the belief to a scalar. It also checks that the model serialization preserves exact probabilities and that an observed repair-success event changes the posterior when success probabilities differ by environment. These tests guard against two implementation shortcuts that would invalidate the stated model: conditioning only on an observation token while ignoring the visible successor, and replacing the environment belief by a single adequacy probability.

I Checked aggregation and multi-slot actions

Proposition I.1 (A sufficient environment partition). *Suppose environments in each block have identical current labels, terminal costs for every declaration, joint event probabilities, and task/direct costs on every state–action–event triple. Then block posterior masses, the visible state, and remaining horizon suffice for optimal Bayes control.*

Proof. Within a block every likelihood and loss vector coordinate is identical. Predictive event probabilities and expected costs therefore depend only on block masses. Summing the individual Bayes update gives Bayes’ rule on these masses. The terminal minimum also depends only on them. Backward induction proves value preservation. This is sufficient and makes no minimality claim. $\square$

The implementation checks these equalities before constructing the quotient; it rejects an invalid proposed merge.

Positive integer macro durations can be expressed without zero-time cycles. Assume an action of duration $d(x, a)$ reveals its joint event at completion, with a specified full-action cost, and cannot be interrupted.

Proposition I.2 (Duration expansion). *If an action is available only when $d(x, a) \leq h$, its macro decision problem is equivalent to a finite unit-slot problem with visible countdown states.*

Proof. Augment each decision state with its remaining horizon. A duration-one action uses the original completion kernel. A longer action first enters a countdown state; intermediate transitions reveal only deterministic time advancement, cost zero, and permit only continuation. Its last transition uses the original joint event and charges its full cost exactly once. Filter actions that cannot finish before the deadline. Every macro policy gives a unit policy by inserting forced continuations, and every unit policy induces the corresponding macro choices. The joint law at decision times, total costs, and terminal declaration law coincide. Therefore their optimal values coincide. $\square$

The adapter uses this completion convention explicitly. It does not assert equivalence when observations arrive during the action or interruption is allowed.

J Finite count-lattice recurrence

For a Bernoulli pre mean p_i , let $w_i(n, u)$ be active mass at count (n, u) before the eligibility check. Initialize $w_i(0, 0) = 1$. Remove selected mass, then propagate each unresolved cell by

$$w_i(n + 1, u) += (1 - p_i)w_i(n, u), \quad w_i(n + 1, u + 1) += p_i w_i(n, u).$$

At $n = N$ check eligibility first and charge every remaining unresolved cell. The policy selects a decoder using all candidate likelihoods; the index i in this recurrence is used only by the evaluator to compute that common policy’s performance.

After selection at n , success on attempt $r \leq R$ has probability $q(1 - q)^{r-1}$ and leaves $H - n - r$ post observations. For a post mean v_i , sum the binomial probabilities whose nearest class mean maps to the wrong old label under the selected decoder. Midpoint ties use the lowest original class index. The implementation accumulates small tails directly from a large term using a probability-mass recurrence; subtraction from one is avoided where it would erase a small tail. These are finite floating-point sums, not certified interval bounds. Their normalization residual is stored separately.

K Original-fixture sensitivity table

Every row changes only its named parameter from the three-environment current-label fixture. The fixed-confidence theorem is not applied to the rows with repair success zero.

Parameter	Setting	Joint	Gate	No repair	Action
repair_cost	0	0.482416	0.834144	0.834144	repair
repair_cost	0.1	0.607344	0.834144	0.834144	repair
repair_cost	0.2	0.731462	0.834144	0.834144	repair
repair_cost	0.4	0.834144	0.834144	0.834144	verify
repair_cost	0.8	0.834144	0.834144	0.834144	verify
repair_success	0	0.834144	0.834144	0.834144	verify
repair_success	0.2	0.834144	0.834144	0.834144	verify
repair_success	0.5	0.834144	0.834144	0.834144	verify
repair_success	0.8	0.731462	0.834144	0.834144	repair
repair_success	1	0.638146	0.834144	0.834144	repair
verify_accuracy	0.5	0.731703	0.920000	0.920000	repair
verify_accuracy	0.65	0.731703	0.920000	0.920000	repair
verify_accuracy	0.8	0.731462	0.834144	0.834144	repair
verify_accuracy	0.95	0.601000	0.601000	0.601000	verify
verify_accuracy	1	0.520000	0.520000	0.520000	verify
horizon	1	0.320000	0.320000	0.320000	fallback
horizon	2	0.440000	0.440000	0.440000	fallback
horizon	4	0.658400	0.658400	0.658400	verify
horizon	6	0.731462	0.834144	0.834144	repair
horizon	8	0.773621	0.983863	0.983863	repair

L Code map and reproduction

The project is self-contained. The complete release includes runnable code and results; the separate arXiv source archive contains the LaTeX, bibliography, and generated figures/tables required to compile this document.

File within <code>src/m2fh/</code>	Purpose
<code>model.py</code> , <code>belief.py</code>	Exact finite model and joint-event Bayes update
<code>solver.py</code> , <code>policies.py</code>	Control, policy restrictions, diagnostics and oracle
<code>enumerator.py</code>	Independent deterministic policy trees
<code>certification.py</code>	Bernoulli fixed-confidence finite protocol
<code>decoder_structure.py</code>	Factorized coefficients and polynomial race
<code>deadline.py</code>	Finite information lower bound and shrinking family
<code>pdf_fixture.py</code> , <code>families.py</code>	Explicit original and binary models
<code>aggregation.py</code> , <code>durations.py</code>	Checked quotient and macro-action expansion
<code>replay.py</code> , <code>cli.py</code>	Synthetic visible-event interface and command line

Run `python3 scripts/reproduce.py` from the project root after installing the declared Python and TeX dependencies. Use `-skip-pdf` explicitly when only computations are wanted. The script writes stage logs, a validation report, source hashes, numerical tables, figures, and the final PDF. `python3 scripts/package_release.py` builds both release archives and a SHA-256 file inventory. The generated tables in the arXiv archive permit compilation without running Python.

The original supplied archive’s output CSVs are preserved as reference data in the provenance directory. They are not presented as independently inspected original Python code. All results in the integrated manuscript are regenerated by the included implementation.